

FINITE-STATE DETERMINISTIC VALIDATION OF THE COLLATZ CONJECTURE

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ABSTRACT. The Collatz conjecture posits that any positive integer x , when repeatedly subjected to two operations:— $3x+1$ when x is odd and $x/2$ when x is even—ultimately reaches 1 without exception. Universality of this conjecture has not been validated despite extensive studies since its official introduction in 1937. This paper develops a finite-state deterministic framework that characterizes the behavior of all Collatz trajectories under modular decomposition. By formulating the iteration process over residue transitions modulo powers of two, the work establishes a closed deterministic system that rules out both potential counterarguments to the conjecture: the existence of non-trivial cycles and the possibility of divergent trajectories. The analysis combines modular recurrence identities, Diophantine constraints, and valuation dynamics to demonstrate that every admissible trajectory must terminate in the trivial loop $(1 \rightarrow 4 \rightarrow 2 \rightarrow 1)$. The framework thus provides a complete validation of the Collatz conjecture within a formally bounded, number-theoretic structure. The entire logical argument has been formally verified in the Lean 4 theorem prover.

1. INTRODUCTION

The Collatz conjecture, stated by Lothar Collatz in 1937, concerns the behavior of the function $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ defined as:

$$f(n) = \begin{cases} 3n + 1 & \text{if } n \equiv 1 \pmod{2} \\ \frac{n}{2} & \text{if } n \equiv 0 \pmod{2} \end{cases}$$

The conjecture asserts that for any starting integer n , the sequence of iterates $f^k(n)$ will eventually reach the integer 1. Despite its elementary statement, a formal proof has remained one of the most elusive problems in mathematics.

1.1. Notable prior works.

- (1) Research on the Collatz conjecture has progressed along two main fronts: massive computational verification and the development of deep theoretical arguments. Computationally, the conjecture has been verified for all integers up to an enormous threshold. Wei Ren [6] conducted a significant brute force experiment related to the Collatz conjecture, focusing on verifying the conjecture for extremely large numbers. He introduced new algorithms capable of verifying numbers of about 100,000 bits (around 30,000 digits), far larger than previous efforts which had reached numbers around 60 bits. His method converted the standard numerical computations involved in Collatz sequences into bit computations, allowing verification for much larger integers without practical upper bounds. Notably, Ren verified that the number $2^{100000} - 1$

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returns to 1 after 481,603 steps of the $3x + 1$ operation and 863,323 steps of halving operations, confirming the Collatz conjecture for this very large number.

- (2) The theoretical investigation, which seeks a general proof, was formalized by early structural approaches. The work of Riho Terras ([9] on parity vectors and the extensive 2-adic analysis by Jeffrey Lagarias [4] created the formal language necessary to analyze the conjecture's two main challenges: non-trivial cycles and divergent trajectories.
- (3) Regarding non-trivial cycles, the most significant results come from Diophantine approximation techniques. The work of Shalom Eliahou [1] established that any such cycle must be astronomically long, with a length of at least 17 billion.
- (4) Regarding divergent trajectories, the main results have been probabilistic. This line of research culminated in the work of Terence Tao [8] who proved that 'almost all' Collatz orbits attain values that are arbitrarily small relative to their starting point, effectively showing that divergent trajectories have a natural density of zero.
- (5) The current landscape, therefore, is one where any potential counterexample has been pushed into the realm of the astronomically large and statistically insignificant. The conjecture remained open because these powerful but incomplete results had not yet closed the final logical gap.

1.2. Theoretical overview.

1.2.1. *The proof strategy.* The proof proceeds by demonstrating the structural impossibility of the two potential counterarguments to the conjecture: non-trivial integer cycles and divergent trajectories. Each counterargument is refuted using a dedicated analytical framework.

- **Part 1: Disproving Non-Trivial Cycles.** The work of Eliahou (1993) used Diophantine inequalities to establish an immense lower bound for any potential cycle. Our Perturbation Model operates on a different axis; instead of bounding a cycle's properties, it proves that the valuation structure required by the cycle equation is inherently contradictory, making a cycle impossible at *any* length.

This framework analyzes all possible cycle structures as deviations from a baseline 'equilibrium state' that corresponds to the trivial $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ loop. The proof demonstrates that any valid perturbation introduces an irreconcilable conflict between the algebraic constraints required for a cycle's integer ratio and the arithmetic constraints imposed by its 3-adic valuation, rendering such a cycle impossible.

- **Part 2: Disproving Divergent Trajectories.** The landmark result of Tao [8] shows that 'almost all' orbits are bounded, a powerful probabilistic statement. However, this does not preclude the existence of a single, exceptional divergent trajectory of measure zero.

The Modular Loop Framework presented here provides a deterministic, encoded and strictly periodic path for every integer if not converged earlier. We prove that a positive integer $2^v \cdot k + m$ with fixed v and bounded m requires indefinite growth of the core integer k for a divergent trajectory - which creates a domain conflict leading to impossibility.

1.2.2. *Summary of lemmas for disproving non-trivial integer cycle.*

TABLE 1. Summary of Cycle Lemmas (Part 1)

Level	Component	Description	Deps
I. Frame-work	Lemma 1A	Establishes the equilibrium state (trivial cycle) and defines cycles as perturbations.	(None)
II. Outcomes	Lemma 1B	Formalizing ΔN (Net deviation).	Lemma 1A
III. Elimination	Lemma 1D	Proves pure positive perturbations force $D_{\text{new}} > N_{\text{new}}$.	Lemma 1A
	Lemma 1C	Proves pure negative perturbations force $N_{\text{new}} > D_{\text{new}}$.	Lemma 1A
IV. Conflict	Lemma 1E	Proves negative perturbations create an Arithmetic Problem (low valuation).	Lemma 1A
	Lemma 1H	Defines the specific positive perturbations required to solve the Arithmetic Problem.	Lemma 1E
V. Contradiction	Lemmas 1F & 1G	Proves the minimal positive perturbations required for Lemma 1H cause fatal algebraic failure.	Lemma 1H
VI. Proof	Theorem 1	Synthesizes the contradiction: Arithmetic and Algebraic conditions are mutually exclusive.	All

Part 1: The Non-Existence of Non-Trivial Integer Cycles. This lemma serves as the logical foundation and theoretical preamble to this paper.

Part 2: The Impossibility of Divergent Trajectories

This section proves that no integer can grow without bound.

TABLE 2. Logical Dependency Mapping for the Disproof of Divergence

Component	Level	Logical Function	Prerequisites
Lemma 2A	I. Foundation	Establishes the formal equivalence between Collatz sequence dynamics and linear Diophantine equations.	None
Lemma 2B	II. Analysis	Defines the trajectory as a deterministic finite-state machine over a constant modulus 2^v .	2A
Lemma 2C	II. Analysis	Proves that path information is rigidly encoded within the 2-adic residues of the core integer k_1 .	2B
Lemma 2D	III. Engine	Derives the linear recurrence relation $n' = \frac{3^L}{2^K}n + \mathcal{C}$ for any closed periodic loop.	2B, 2C
Lemma 2E	III. Engine	Generalizes multiplier products and additive tails for heterogeneous loop transitions.	2D
Lemma 2F	IV. Contradiction	Proves the 2-adic limit K_∞ of any infinite path lies outside the integer domain $\mathbb{Z}^+$.	2C, 2E
Theorem 2	V. Synthesis	Formalizes Bit-Exhaustion: the finite capacity of n_0 forces a monotonic collapse of growth trajectories.	All

2. THEORY

2.1. Modular class analysis under Collatz operations. There are 2^n residue classes modulo 2^n . Since exactly half of the integers in $\{0, 1, \dots, 2^n - 1\}$ are odd, the number of odd residue classes modulo 2^n is 2^{n-1} .

The odd modular classes can be written as: $2^n \cdot k + m$ (core integer $k \in \mathbb{Z}^+$, and residue $m \in \{1, 3, 5, \dots, 2^n - 1\}$).

Let $N(a, n)$ be the number of odd residue classes m modulo 2^n such that the 2-adic valuation $v_2(3m + 1)$ is exactly a . For positive integers $a \leq n$, this number is given by:

Lemma 0. *Let $N(a, n)$ be the number of odd residue classes m modulo 2^n such that the 2-adic valuation $v_2(3m + 1)$ is exactly a . For positive integers $a \leq n$, this number is given by:*

$$N(a, n) = \begin{cases} 2^{n-a-1} & \text{if } 1 \leq a < n \\ 1 & \text{if } a = n \quad (\text{Edge case } v_2 \geq n) \end{cases}$$

Proof. Let $N = 2^n \cdot k + m$, then $3N + 1 = 3 \cdot 2^n \cdot k + 3m + 1$.

$3 \cdot 2^n$ is divisible by 2^n .

Step I: Existence and Uniqueness: Since $\gcd(3, 2^a) = 1$, the congruence $3m + 1 \equiv 0 \pmod{2^a}$ has a unique solution modulo 2^a . Let this solution be $m \equiv c \pmod{2^a}$.

Step II: Lifting solution to mod 2^n :

Any solution $m \equiv c \pmod{2^a}$ can be expressed as $m = c + 2^a \cdot k$ for some integer k . To find all solutions modulo 2^n , we consider $k \pmod{2^{n-a}}$. There are 2^{n-a} distinct values of $k \pmod{2^n}$ leading to 2^{n-a} residues $m \pmod{2^n}$.

Step III: Excluding higher divisibility:

To ensure $3m + 1 \equiv 0 \pmod{2^n}$, we exclude solutions where $m \equiv c \pmod{2^n}$. Each solution modulo 2^n splits into two cases modulo 2^{a+1} :

- a) One satisfies $3m + 1 \equiv 0 \pmod{2^{a+1}}$
- b) The other does not satisfy $3m + 1 \equiv 0 \pmod{2^{a+1}}$

One half of the lifted solutions, i.e., 2^{n-a-1} remain valid for exact divisibility 2^a .

Step IV: Counting for each:

For $1 \leq a \leq n$: The number of residues $m \pmod{2^n}$ with $v_2(3m + 1) = a$ is 2^{n-a-1} .

Special edge case for $a = n$: There exactly 1 residue $m \pmod{2^n}$ such that $(3m + 1) \equiv 0 \pmod{2^n}$. $\square$

Corollary 0-A. *According to Lemma 0, applying the $3x + 1$ operation to the set of odd modular classes yields the following distribution of division exponents:*

- 2^{n-2} results have divisibility by 2^1 (division exponent $a = 1$)
- 2^{n-3} results have divisibility by 2^2 (division exponent $a = 2$)
- 2^{n-4} results have divisibility by 2^3 (division exponent $a = 3$)
- 2^{n-5} results have divisibility by 2^4 (division exponent $a = 4$)
- $\vdots$
- $2^{n-n} = 1$ result has divisibility by 2^n (division exponent $a = n$)

The highest possible value of the division exponent $a = v_2(3x + 1)$ also depends on the parity of k in the expression $3 \cdot 2^n \cdot k + 3m + 1$. After factoring out 2^n , the term becomes $3k + \frac{3m+1}{2^n}$. Since $3m + 1$ is always even, the parity of the entire expression depends on

whether $3k$ is even or odd — which is determined by the parity of k . If k is odd, then $3k$ is also odd, and adding it to the (odd) quotient $\frac{3m+1}{2^n}$ yields an even result. Thus, the parity of k influences whether an additional factor of 2 can be extracted, affecting the total valuation $v_2(3x+1)$. Therefore, the largest exponent (say, a_k) modulo 2^v is k -dependent such that $a_k \geq v$.

Corollary 0-B: Finite Paths. *In the Collatz sequence, one odd modular class transforms into another odd modular class within the premises of $2^v \cdot k + m$, forming a finite-state transition system. Between finite states, there exists a finite number of transformation paths from each modular state to any other state.*

Corollary 0-C. *The higher the moduli, the greater the division exponents.*

Corollary 0-D. *Half of the exponents in the whole modular landscape are growth-inducing (exponents = 1) and the remaining half are contraction-inducing. The fact that the effect of the contraction-inducing half is greater than the growth-inducing half establishes an inherent bias towards convergence.*

3. MULTIPLE ITERATIONS ON MODULAR CLASSES

We define the specific Collatz iteration terms as follows:

- (a) I^1 represents a single $3x+1$ operation on a starting odd integer N_0 .
- (b) I^2 represents division by 2^a , where $a = v_2(3N_0+1)$.
- (c) I^d represents a single set of I^1 followed by an I^2 operation.

Let us assume a sequence with n allowed transformations in which the divisional exponents involved are $a_1, a_2, a_3, \dots, a_n$.

The operation I^d applied to $2^v \cdot k_1 + m_1$ yields successive odd integers as follows:

Step 1:

$$2^v k_1 + m_1 \rightarrow \frac{3 \cdot 2^v \cdot k_1 + 3m_1 + 1}{2^{a_1}}$$

Step 2:

$$\frac{3 \cdot 2^v k_1 + 3m_1 + 1}{2^{a_1}} \rightarrow \frac{3^2 \cdot 2^v \cdot k_1 + 3^2 \cdot m_1 + 3 + 2^{a_1}}{2^{a_1+a_2}}$$

Step 3:

$$\dots \rightarrow \frac{3^3 2^v k_1 + 3^3 \cdot m_1 + 3^2 + 3 \cdot 2^{a_1} + 2^{a_1+a_2}}{2^{a_1+a_2+a_3}}$$

Step 4:

$$\dots \rightarrow \frac{3^4 2^v \cdot k_1 + 3^4 \cdot m_1 + 3^3 + 3^2 \cdot 2^{a_1} + 3 \cdot 2^{a_1+a_2} + 2^{a_1+a_2+a_3}}{2^{a_1+a_2+a_3+a_4}}$$

Step 5:

$$\dots \rightarrow \frac{3^5 \cdot 2^v \cdot k_1 + 3^5 \cdot m_1 + 3^4 + \dots + 3 \cdot 2^{a_1+a_2+a_3} + 2^{a_1+\dots+a_4}}{2^{a_1+\dots+a_5}}$$

For such a sequence with n steps ($1 \leq n \leq 2^{v-1}$), if the integer becomes $2^v \cdot k_{n+1} + m_{n+1}$, then we arrive at the **Fundamental Linear Diophantine Equation** of the Collatz conjecture:

$$2^v \cdot k_{n+1} + m_{n+1} =$$

$$\frac{3^n \cdot 2^v \cdot k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + a_2 + \dots + a_n}}$$

Now, if $2^v \cdot k_{n+1} + m_{n+1} = 2^v \cdot k_1 + m_1$ (i.e., an integer reappears in a Collatz sequence after arbitrary n iterations of I^d), then by the equation above:

$$2^v \cdot k_1 + m_1 =$$

$$\frac{3^n \cdot 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + a_2 + \dots + a_n}}$$

This is the fundamental linear Diophantine equation representing an integer loop.

Lemma 1A. *If all division exponents $a_i = p$, the only possible integer cycle is $4 \rightarrow 2 \rightarrow 1 \rightarrow 4$ when $p = 2$.*

Proof. Notation and setup:

- Let n be the length of a hypothetical cycle, an integer where $n \geq 1$.
- Let $a = (a_1, \dots, a_n)$ be the vector of division exponents, where each $a_i \in \mathbb{Z}^+$.
- Let $S_j = \sum_{i=1}^j a_i$ be the cumulative sum of the first j exponents, with the convention that $S_0 = 0$.
- Let $S = \sum_{i=1}^n a_i$ be the total sum of all exponents.
- The Numerator (N) of the cycle ratio is defined as: $N = N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$.
- The Denominator (D) of the cycle ratio is defined as: $D = 2^S - 3^n$.
- The Cycle Ratio (R) is defined as $R = \frac{N}{D}$.
- The Equilibrium State is defined as the special arrangement when all division exponents $a_i = 2$, $N = N_{\text{eq}}$, $D = D_{\text{eq}}$ and $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$. This implies $R_{\text{eq}} = 1$.

Recalling the fundamental equation, writing $S = \sum_{i=1}^n a_i$, and $S_j = \sum_{i=1}^{n-1} a_i$:

$$2^v \cdot k_1 + m_1 =$$

$$\frac{3^n 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j}}{2^S}$$

Rearranging terms:

$$\begin{aligned} 2^v \cdot k_1 \cdot 2^S + m_1 \cdot 2^S &= 3^n 2^v \cdot k_1 + 3^n \cdot m_1 + 3^{n-1} \\ &\quad + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j} \end{aligned}$$

Grouping the k_1 and m_1 terms:

$$\begin{aligned} 2^v \cdot k_1 (2^S - 3^n) &= m_1 \cdot (3^n - 2^S) + 3^{n-1} \\ &\quad + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j} \end{aligned}$$

Solving for k_1 :

$$\begin{aligned} k_1 &= \frac{m_1(3^n - 2^S) + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j}}{(2^S - 3^n)} \cdot \frac{1}{2^v} \\ &= -\frac{m_1}{2^v} + \frac{3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j}}{2^v \cdot (2^S - 3^n)} \\ &= \frac{1}{2^v} [R - m_1] \end{aligned}$$

Where the ratio R is defined as:

$$R = \frac{3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{S_j}}{(2^S - 3^n)} = \frac{N}{D}$$

Applying the Equilibrium Condition: The lemma considers the equilibrium state, where all division exponents are equal to a constant integer p , i.e., $a_i = p$ for all $i = 1, \dots, n$. Under this condition, the cumulative sums S_j take a simple, regular form:

- $S_0 = 0$
- $S_1 = p$
- $S_2 = p + p = 2p$
- In general, for any $j \geq 0$, the cumulative sum is $S_j = j \cdot p$.

Formulating the Numerator Sum: Substituting this into the general formula for the numerator gives the equilibrium numerator, N_{eq} :

$$N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{j \cdot p}.$$

Demonstrating the Geometric Progression: To show that this sum is a Geometric Progression (GP), we write out the first few terms:

- For $j = 0$: The term is $3^{n-1} \cdot 2^{0 \cdot p} = 3^{n-1}$.
- For $j = 1$: The term is $3^{n-2} \cdot 2^{1 \cdot p} = 3^{n-2} \cdot 2^p$.
- For $j = 2$: The term is $3^{n-3} \cdot 2^{2 \cdot p} = 3^{n-3} \cdot (2^p)^2$.

The sum can be written as: $N_{\text{eq}} = 3^{n-1} + 3^{n-2} \cdot 2^p + 3^{n-3} \cdot (2^p)^2 + \dots + 3^0 \cdot (2^p)^{n-1}$.

Considering any j -th term, $T_j = 3^{n-1-j} \cdot (2^p)^j$, the ratio of consecutive terms is constant:

$$\frac{T_{j+1}}{T_j} = \frac{2^p}{3}$$

Since this ratio is constant for all terms, the sum is a Geometric Progression with the following parameters:

- **First Term (A):** 3^{n-1}
- **Common Ratio (r):** $\left(\frac{2^p}{3}\right)$
- **Number of Terms:** n

Summing the Series and Finalizing the Proof:

We apply the standard formula for the sum of a finite geometric series, $\text{sum} = A \cdot \left[\frac{r^n - 1}{r - 1}\right]$:

$$N_{\text{eq}} = 3^{n-1} \cdot \left[\frac{\left(\frac{2^p}{3}\right)^n - 1}{\frac{2^p}{3} - 1} \right] \implies N_{\text{eq}} = \frac{2^{np} - 3^n}{2^p - 3}$$

The equilibrium denominator is $D_{\text{eq}} = 2^S - 3^n = 2^{np} - 3^n$. The cycle ratio R is therefore:

$$R = \frac{N}{D} = \left[\frac{2^{n \cdot p} - 3^n}{2^p - 3} \right] \cdot \left[\frac{1}{2^{n \cdot p} - 3^n} \right] \Rightarrow R = \frac{1}{2^p - 3}$$

For R to be a positive integer, the denominator $2^p - 3$ must equal 1. This occurs only when $2^p = 4$, which implies $p = 2$. In this case, $R = 1$. Substituting into the equation for the core integer, $k_1 = \frac{1}{2^v} \cdot [R - m_1]$, gives $k_1 = \frac{1}{2^v} \cdot [1 - m_1]$. Since k_1 must be a non-negative integer and m_1 is a positive odd residue, the only solution is $m_1 = 1$, which yields a valid $k_1 = 0$. This corresponds to the integer 1, proving that the only possible cycle with uniform exponents is the trivial one. $\square$

Corollary 1A-1: The Threshold Ratio. *For R to be a positive integer, the denominator $D = 2^S - 3^n$ must be greater than 0. The threshold value for which $2^S - 3^n > 0$ is $2^S > 3^n$, or $\frac{S}{n} > \log_2 3$. This gives us the threshold ratio:*

$$\frac{S}{n} > 1.585, \quad \text{or} \quad S > 1.585 \cdot n$$

4. THE PERTURBATION MODEL

By Lemma 1A, it is evident that no integer solution is possible for R unless all division exponents $a_i = 2$. Assuming this equilibrium, where $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$, as a base case, all possible combinations of division exponents can be examined by incrementing or decrementing them from this equilibrium (base) position. The model assumes that a non-trivial cycle is formed by some appropriate combination of division exponents that makes $R = \frac{N}{D}$ a positive integer. The following perturbations cover all valid division exponent combinations:

- **Case I:** Pure increment (positive perturbations) of the exponents from a_1 to a_n .
- **Case II:** Pure decrement (negative perturbations) of the division exponents from a_1 to a_n .
- **Case III:** Mixed perturbations with net increment in division exponent sum.
- **Case IV:** Mixed perturbations with net decrement in division exponent sum.
- **Case V:** Mixed perturbations with no net alteration in the division exponent sum (total increment = total decrement).

All cases can be captured in two scenarios: either an increase or a decrease in the numerator's magnitude while keeping $N_{\text{new}} > D_{\text{new}}$. There is, however, another possibility in which, by some exceptional coincidence, $N_{\text{new}} = N_{\text{eq}}$ is achieved via a net decrement of the total division exponents.

Lemma 1B. *Due to valid perturbations of division exponents from equilibrium, the possible outcomes are: (a) $N_{\text{new}} > N_{\text{eq}}$, (b) $N_{\text{new}} < N_{\text{eq}}$, or (c) a non-trivial $N_{\text{new}} = N_{\text{eq}}$.*

Proof. Notation and Setup:

Let $n \geq 1$ be a fixed integer representing the cycle length. The exponent vector is denoted by $a = (a_1, \dots, a_n)$, where each a_i is an integer $a_i \geq 1$. The perturbation of each exponent from the equilibrium value of 2 is defined as $\delta_i = a_i - 2$.

We define the prefix sums of the perturbations as $S'_j = \sum_{i=1}^j \delta_i$ for $j = 0, 1, \dots, n-1$, with the convention that $S'_0 = 0$. The cumulative sum of the exponents themselves is $s_j = \sum_{i=1}^j a_i$. Note the relationship $s_j = 2j + S'_j$.

The numerator N associated with an exponent vector a is defined as:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{s_j}$$

The equilibrium numerator, N_{eq} , corresponds to the equilibrium vector $a_{\text{eq}} = (2, 2, \dots, 2)$.

Exact Difference Formula:

For any perturbation vector $\delta = (\delta_1, \dots, \delta_n)$, the difference between the perturbed numerator and the equilibrium numerator is given by the identity:

$$N_{\text{new}} - N_{\text{eq}} = \sum_{j=1}^{n-1} 3^{n-1-j} 2^{2j} \left[2^{S'_j} - 1 \right]$$

Proof of Formula. We begin with the definition of N_{new} and substitute the relationship $s_j = 2j + S'_j$:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j+S'_j}$$

Factoring the power of 2, we get:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot 2^{S'_j}$$

The equilibrium numerator N_{eq} is the case where all $\delta_i = 0$, which implies $S'_j = 0$ for all j . In this case, $2^{S'_j} = 2^0 = 1$, so:

$$N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j}$$

Subtracting the equilibrium value N_{eq} from N_{new} and factoring out the common term $3^{n-1-j} \cdot 2^{2j}$ yields the identity:

$$\begin{aligned} N_{\text{new}} - N_{\text{eq}} &= \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot 2^{S'_j} - \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \\ &= \sum_{j=1}^{n-1} 3^{n-1-j} \cdot 2^{2j} \cdot \left[2^{S'_j} - 1 \right] \end{aligned}$$

Consequences of the Identity

This identity leads to several immediate and exhaustive consequences for the behavior of the numerator. Let the sensitivity weights be defined as $w_j = 3^{n-1-j} \cdot 2^{2j}$, which are strictly positive for all j .

- (1) **Equality Case** ($N_{\text{new}} = N_{\text{eq}}$): $N_{\text{new}} = N_{\text{eq}}$ if and only if $S'_j = 0$ for every $j = 0, \dots, n-1$. This is equivalent to $\delta_i = 0$ for all i (i.e., the exponent vector is the equilibrium vector).

Proof. The difference $N_{\text{new}} - N_{\text{eq}}$ is a sum of terms $w_j \cdot (2^{S'_j} - 1)$. Since each weight w_j is positive, a term is zero if and only if $2^{S'_j} - 1 = 0$, which occurs if and only if $S'_j = 0$. A sum of terms that are all ≥ 0 or all ≤ 0 can only be zero if every term is zero. Thus, $S'_j = 0$ for all j .

(2) **Monotone Sign Cases:**

- (a) If $S'_j \geq 0$ for every j and at least one $S'_j > 0$, then $N_{\text{new}} > N_{\text{eq}}$. *Proof.* Each factor $(2^{S'_j} - 1)$ is non-negative, and at least one is strictly positive. Since all weights w_j are positive, the total sum must be strictly positive.
- (b) If $S'_j \leq 0$ for every j and at least one $S'_j < 0$, then $N_{\text{new}} < N_{\text{eq}}$. *Proof.* Analogously, each term in the sum is non-positive, and at least one is strictly negative. The total sum must be strictly negative.

(3) **Mixed-Sign Prefixes (General Case):** Based on the identity $\Delta N = N_{\text{new}} - N_{\text{eq}}$, for any non-trivial perturbation, there are three logical possibilities for the relationship between the perturbed numerator N_{new} and the equilibrium numerator N_{eq} :

- (a) $N_{\text{new}} > N_{\text{eq}}$ (The numerator increases).
- (b) $N_{\text{new}} < N_{\text{eq}}$ (The numerator decreases).
- (c) $N_{\text{new}} = N_{\text{eq}}$ (The numerator remains unchanged due to a coincidental cancellation).

□

4.1. **Constraints in perturbation model.** There are some mathematical constraints that create a deterministic frame around this model:

4.1.1. *Constraint 1: Division Exponent Lower Bound.* The division exponent is defined as $a_i = v_2(3x_{i-1} + 1)$. We note that since x_{i-1} is a positive odd integer, $(3x_{i-1} + 1)$ is a positive even integer, which implies the constraint that $a_i \geq 1$ for all i . Consequently, the only possible negative perturbation from the equilibrium value of 2 is $\delta_i = -1$, i.e., **no division exponent can be lowered more than once.**

4.1.2. *Constraint 2: Modulo 3 Incompatibility.* Both D and N at any form (perturbed or unperturbed) are odd and not divisible by 3.

4.1.3. *Constraint 3: Ratio Bounds.* If an integer cycle forms, the ratio $R_{\text{new}} = \frac{N_{\text{new}}}{D_{\text{new}}} \geq 5$ for all $N_{\text{new}} > D_{\text{new}}$ cases, and $\frac{N_{\text{new}}}{D_{\text{new}}} = 1$ for some non-trivial $N_{\text{new}} = D_{\text{new}}$.

Lemma 1C (Pure-Negative Perturbations Force $N_{\text{new}} > D_{\text{new}}$). *Let us assume $n \geq 1$ and let the exponent-vector $a = (a_1, \dots, a_n)$ satisfy $1 \leq a_i \leq 2$ for every i . Let us also assume that the perturbation is non-trivial, meaning at least one $a_i < 2$ (and thus $a_i = 1$). Let $S = \sum_{i=1}^n a_i$ be the total sum of exponents. The negatively perturbed numerator and denominator are defined as: $N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S_j}$ and $D_{\text{new}} = 2^S - 3^n$. Then, under these conditions, it is always true that $N_{\text{new}} > D_{\text{new}}$.*

Proof. The proof proceeds by establishing a tight lower bound for the numerator, N_{new} , and then showing that this lower bound is strictly greater than the denominator, D_{new} .

- (1) **Establishing a Lower Bound for the Cumulative Sums (S_j):** For any index j in the range $0 \leq j \leq n-1$, the portion of the total sum of exponents *after* index j is $S - S_j = \sum_{i=j+1}^n a_i$. Since each $a_i \leq 2$, this sum is bounded above by the number of terms multiplied by 2:

$$S - S_j = \sum_{i=j+1}^n a_i \leq 2(n - j)$$

Rearranging this inequality gives a lower bound for the cumulative sum S_j :

$$S_j \geq S - 2(n - j)$$

- (2) **Establishing a Lower Bound for the Numerator (N_{new}):** We insert this lower bound for S_j into the definition of the numerator:

$$N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S_j} \geq \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S-2(n-j)}$$

Factoring out the term 2^S :

$$N_{\text{new}} \geq 2^S \cdot \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{-2(n-j)}$$

To simplify the sum formula, let us introduce $m = n - j$ so that:

$$N_{\text{new}} \geq 2^S \cdot \sum_{m=1}^n 3^{m-1} \cdot 4^{-m}$$

The sum component is a Geometric Progression:

$$\begin{aligned} \sum_{m=1}^n 3^{m-1} \cdot 4^{-m} &= \frac{1}{4} \sum_{m=1}^n \frac{3^{m-1}}{4^{m-1}} \\ &= \frac{1}{4} \sum_{m=1}^n \left(\frac{3}{4}\right)^{m-1} \end{aligned}$$

This is a GP with first term $A = 1$, common ratio $r = \frac{3}{4}$, and number of terms $= n$. Hence:

$$\begin{aligned} \frac{1}{4} \sum_{m=1}^n \left(\frac{3}{4}\right)^{m-1} &= \frac{1}{4} \left[\frac{1 - \left(\frac{3}{4}\right)^n}{1 - \frac{3}{4}} \right] \\ &= \frac{1}{4} \left[\frac{1 - \left(\frac{3}{4}\right)^n}{\frac{1}{4}} \right] \\ &= 1 - \left(\frac{3}{4}\right)^n \end{aligned}$$

Substituting this back, we obtain the lower bound for N_{new} :

$$N_{\text{new}} \geq 2^S \left[1 - \left(\frac{3}{4}\right)^n \right]$$

- (3) **Comparing the Numerator and Denominator:** We now analyze the difference $N_{\text{new}} - D_{\text{new}}$ using this lower bound:

$$\begin{aligned} N_{\text{new}} - D_{\text{new}} &\geq 2^S \cdot \left[1 - \left(\frac{3}{4} \right)^n \right] - (2^S - 3^n) \\ &\geq 2^S - 2^S \left(\frac{3}{4} \right)^n - 2^S + 3^n \\ &\geq 3^n - \left(\frac{3}{4} \right)^n \cdot 2^S \end{aligned}$$

- (4) **Finalizing the Inequality:** Let p be the number of negative perturbations (i.e., the number of exponents a_i that are equal to 1). By the lemma's premise, $p \geq 1$. The total sum of the exponents S is therefore exactly $S = 2n - p$. We start from the inequality already derived in the proof:

$$N_{\text{new}} - D_{\text{new}} \geq 3^n - \left(\frac{3}{4} \right)^n \cdot 2^S$$

Now, we substitute the exact value for S :

$$N_{\text{new}} - D_{\text{new}} \geq 3^n - \left(\frac{3}{4} \right)^n \cdot 2^{2n-p}$$

Since $4^n = (2^2)^n = 2^{2n}$, we can simplify:

$$\begin{aligned} N_{\text{new}} - D_{\text{new}} &\geq 3^n - \left(\frac{3^n}{2^{2n}} \right) \cdot 2^{2n} \cdot 2^{-p} \\ N_{\text{new}} - D_{\text{new}} &\geq 3^n - 3^n \cdot 2^{-p} \end{aligned}$$

Factoring out 3^n gives us the explicit numeric bound:

$$N_{\text{new}} - D_{\text{new}} \geq 3^n \left(1 - \frac{1}{2^p} \right)$$

Conclusion: Since $p \geq 1$, the term $\frac{1}{2^p}$ is always positive and less than or equal to $\frac{1}{2}$. Therefore, the term $(1 - \frac{1}{2^p})$ is always positive (specifically ≥ 0.5). As 3^n is also positive, their product is strictly greater than 0. This proves conclusively that $N_{\text{new}} > D_{\text{new}}$. □

Lemma 1D. *Any purely positive perturbation of the division exponents from the equilibrium leads to a non-integer ratio $\frac{N_{\text{new}}}{D_{\text{new}}}$, violating the cycle condition.*

Proof. **Notation and Setup:**

- $a = (a_1, \dots, a_n)$: The vector of perturbed division exponents.
- $\delta_i = a_i - 2$: The perturbation for each exponent. For this lemma, all $\delta_i \geq 0$, and at least one $\delta_i > 0$.
- $S'_j = \sum_{i=1}^j \delta_i$: The prefix sum of the first j perturbations. By convention, $S'_0 = 0$.
- $S'_n = \sum_{i=1}^n \delta_i$: The total sum of all perturbations. By the lemma's premise, $S'_n \geq 1$.
- $N_{\text{eq}}, D_{\text{eq}}$: The numerator and denominator at the equilibrium state ($a_i = 2$). We know $N_{\text{eq}} = D_{\text{eq}} = 2^{2n} - 3^n$.

- $N_{\text{new}}, D_{\text{new}}$: The perturbed numerator and denominator after applying the perturbations.
- $\Delta N = N_{\text{new}} - N_{\text{eq}}$ and $\Delta D = D_{\text{new}} - D_{\text{eq}}$: The net change in the numerator and denominator, respectively.

The proof proceeds in four steps. We will show that the increase in the denominator (ΔD) is strictly greater than the increase in the numerator (ΔN).

Step 1: The Objective. Our goal is to prove that $D_{\text{new}} > N_{\text{new}}$. Since $D_{\text{eq}} = N_{\text{eq}}$ at equilibrium, this is equivalent to proving that the change in the denominator is strictly greater than the change in the numerator:

$$\Delta D > \Delta N$$

Step 2: Calculating the Exact Change in the Denominator (ΔD). The total sum of the perturbed exponents is $S = 2n + S'_n$. The new denominator is $D_{\text{new}} = 2^S - 3^n = 2^{2n+S'_n} - 3^n$.

We calculate ΔD directly:

$$\begin{aligned} \Delta D &= D_{\text{new}} - D_{\text{eq}} \\ &= (2^{2n+S'_n} - 3^n) - (2^{2n} - 3^n) \\ &= 2^{2n} \cdot 2^{S'_n} - 2^{2n} \end{aligned}$$

Factoring out the 2^{2n} term gives the exact change:

$$\Delta D = 2^{2n} (2^{S'_n} - 1)$$

Step 3: Establishing a strict upper bound for ΔN . We use the Exact Difference Formula from Lemma 1B:

$$\Delta N = \sum_{j=1}^{n-1} \left[3^{n-1-j} 2^{2j} (2^{S'_j} - 1) \right]$$

Since this is a purely positive perturbation ($\delta_i \geq 0$), the sequence of prefix sums is non-decreasing: $0 = S'_0 \leq S'_1 \leq \dots \leq S'_{n-1} \leq S'_n$. This means that for any prefix sum S'_j in the sequence, $S'_j \leq S'_n$. Consequently:

$$(2^{S'_j} - 1) \leq (2^{S'_n} - 1)$$

We can now establish an upper bound for ΔN by replacing each term in the sum with this maximum factor:

$$\Delta N \leq \sum_{j=1}^{n-1} \left[3^{n-1-j} 2^{2j} (2^{S'_n} - 1) \right]$$

Since $(2^{S'_n} - 1)$ is constant with respect to j , we factor it out:

$$\Delta N \leq (2^{S'_n} - 1) \sum_{j=1}^{n-1} [3^{n-1-j} 2^{2j}]$$

The sum on the right is the exact definition of the equilibrium numerator, N_{eq} . This gives us the strict upper bound:

$$\Delta N \leq (2^{S'_n} - 1) \cdot N_{\text{eq}}$$

Step 4: Comparison and Conclusion. We compare the two results:

$$(1) \Delta D = (2^{S'_n} - 1) \cdot 2^{2n}$$

$$(2) \Delta N \leq (2^{S'_n} - 1) \cdot N_{\text{eq}}$$

By definition, $N_{\text{eq}} = 2^{2n} - 3^n$. Since $n \geq 1$, $3^n > 0$, which implies $N_{\text{eq}} < 2^{2n}$. Since the lemma requires at least one positive perturbation, $S'_n \geq 1$, the factor $(2^{S'_n} - 1)$ is a positive number.

It follows directly that:

$$(2^{S'_n} - 1) \cdot N_{\text{eq}} < (2^{S'_n} - 1) \cdot 2^{2n}$$

Substituting our terms back, we get $\Delta N < \Delta D$.

Since the increase in the denominator is strictly greater than the increase in the numerator, it is proven that $D_{\text{new}} > N_{\text{new}}$. Therefore, no purely positive perturbation can satisfy the condition for an integer cycle. $\square$

4.1.4. *Constraint 4: Formulation of Actual ΔN .* The sum formula of the perturbed numerator is:

$$N_{\text{new}} = 3^{n-1} + 3^{n-2}2^{S_1} + 3^{n-3}2^{S_2} + \dots + 2^{S_{n-1}}$$

where $S_1, S_2, \dots, S_{n-1}$ are the perturbed prefix sums of exponents.

The unperturbed (equilibrium) numerator is:

$$N_{\text{eq}} = 3^{n-1} + 3^{n-2} \cdot 2^2 + 3^{n-3} \cdot 2^4 + \dots + 2^{2n-2}$$

If perturbations cause $N_{\text{eq}} > N_{\text{new}}$, then $\Delta N_{\text{actual}} = N_{\text{eq}} - N_{\text{new}}$. In this difference, all unperturbed terms cancel out, leaving only the sum of differences at perturbed positions. If a perturbation is introduced at the k -th term, the actual change is:

$$\Delta N_{\text{actual}} = \sum_{k=2}^n (\text{Term}_{\text{eq},k} - \text{Term}_{\text{new},k})$$

where the term at step k involves the sum of exponents up to the previous step S_{k-1} . Specifically:

$$\text{Term}_k = 3^{n-k} \cdot [2^{2k-2} - 2^{S_{k-1}}]$$

Conversely, if $N_{\text{new}} > N_{\text{eq}}$, then $\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}}$, and the general term becomes:

$$\text{Term}_k = 3^{n-k} \cdot [2^{S_{k-1}} - 2^{2k-2}]$$

4.1.5. *Constraint 5: Bounds on Negative Perturbations.* The number of pure negative perturbations p must satisfy $p \geq 1$ and $p < 0.415 \cdot n$. This is derived from the threshold ratio $S > 1.585n$. Since $S = 2n - p$ (in the pure negative case), we have:

$$2n - p > 1.585n \implies p < (2 - 1.585)n \implies p < 0.415n$$

Similarly, any total decrement in a mixed case must maintain this bound to ensure $D_{\text{new}} > 0$.

4.1.6. *Constraint 6: Formulation of Required ΔN .* This equation defines the exact value of the net change in the numerator, $\Delta N_{\text{required}}$, required for a cycle with a specific integer ratio $Z \geq 5$ to exist.

1. Derivation for Net Decrement ($N_{\text{new}} < N_{\text{eq}}$): Start with the cycle condition $Z = \frac{N_{\text{new}}}{D_{\text{new}}}$, which implies $N_{\text{new}} = Z \cdot D_{\text{new}}$. Substituting $N_{\text{new}} = N_{\text{eq}} - \Delta N$:

$$\begin{aligned} N_{\text{eq}} - \Delta N &= Z \cdot D_{\text{new}} \\ \Delta N &= N_{\text{eq}} - Z \cdot D_{\text{new}} \\ &= (2^{2n} - 3^n) - Z(2^S - 3^n) \\ &= (2^{2n} - Z \cdot 2^S) + (Z \cdot 3^n - 3^n) \end{aligned}$$

This yields the final equation for net decrement:

$$\Delta N_{\text{required}} = (2^{2n} - Z \cdot 2^S) + (Z - 1)3^n \quad (1)$$

Correspondingly, the actual change is summed as:

$$\Delta N_{\text{actual}} = \sum_{k=2}^n 3^{n-k} (2^{2k-2} - 2^{S_{k-1}})$$

2. Derivation for Net Increment ($N_{\text{new}} > N_{\text{eq}}$): Substituting $N_{\text{new}} = N_{\text{eq}} + \Delta N$:

$$\begin{aligned} N_{\text{eq}} + \Delta N &= Z \cdot D_{\text{new}} \\ \Delta N &= Z(2^S - 3^n) - (2^{2n} - 3^n) \\ &= (Z \cdot 2^S - 2^{2n}) + (3^n - Z \cdot 3^n) \end{aligned}$$

This yields the final equation for net increment:

$$\Delta N_{\text{required}} = (Z \cdot 2^S - 2^{2n}) + (1 - Z)3^n \quad (2)$$

Correspondingly, the actual change is summed as:

$$\Delta N_{\text{actual}} = \sum_{k=2}^n 3^{n-k} (2^{S_{k-1}} - 2^{2k-2})$$

4.1.7. *The Fundamental Cycle Condition.* According to the perturbation model, an integer cycle is formed if and only if the actual change in the numerator, ΔN_{actual} (derived from the sum of bitwise changes), reconciles perfectly with the theoretically required change, $\Delta N_{\text{required}}$ (derived from the Diophantine ratio Z).

Thus, the fundamental equation to be tested for the existence of any cycle is:

$$\boxed{\Delta N_{\text{required}} = \Delta N_{\text{actual}}}$$

5. MATHEMATICAL CONSEQUENCES OF PERTURBATIONS

Lemma 1E: First Negative-Perturbation Lemma. *Let a perturbation from equilibrium contain at least one negative component ($\delta_i = -1$). Let r be the index of the first such negative perturbation in the sequence. Then the corresponding term T_{r+1} within the sum for ΔN_{actual} has a 3-adic valuation given by $v_3(T_{r+1}) = n - (r + 1)$. Since $r \geq 1$, this valuation is strictly positive and strictly less than n [$0 < v_3(T_{r+1}) < n$].*

Proof. **1. Setup and Hypotheses:**

- Let $a_1, a_2, \dots, a_n$ be a sequence of integers.
- Let $r \geq 1$ be the **first index** for which $a_r \neq 2$. This implies that $a_1 = a_2 = \dots = a_{r-1} = 2$.
- We consider the specific case where this first perturbation is $a_r = 1$.
- Let $S_k = \sum_{i=1}^k a_i$ denote the partial sum of the sequence.

Note that the total valuation of the sum $v_3(\Delta N)$ will be dictated by its lowest-valuation term (the drop). By the Strong Triangle Inequality (or non-Archimedean property), the sum of two terms $v_3(A + B)$ can only exceed $\min(v_3(A), v_3(B))$ if and only if $v_3(A) = v_3(B)$. Since the base valuations of the Collatz terms $(n - 1 - j)$ are strictly staggered, any repair relies on the perturbation factors $v_3(2^d - 1)$ achieving a perfect, non-random alignment.

2. Definition:

- For the index $k = r + 1$, we define a quantity d_k as follows: $d_k = |S_{k-1} - (2k - 2)|$.

3. Claims:

The lemma makes two claims:

- The value d_k is an **odd integer**.
- Consequently, the 3-adic valuation of the expression is precisely $n - k$. Since $k \geq 2$, this valuation is at most $n - 2$:

$$v_3 [3^{n-k} \cdot \{2^{S_{k-1}} - 2^{2k-2}\}] = n - k \leq n - 2$$

Proof of Claims

The proof proceeds by first verifying that d_k is odd and then using this fact to calculate the valuation.

Step 1: Calculating the partial sum at the point of perturbation. By our hypothesis, the sequence begins with $r - 1$ terms equal to 2, followed by the term $a_r = 1$. The partial sum up to index r , denoted S_r , is therefore:

$$S_r = \sum_{i=1}^r a_i = (2 + 2 + \dots + 2) + 1 = 2(r - 1) + 1 = 2r - 1$$

Step 2: Proof that d_k is odd. The lemma defines d_k for $k = r + 1$. Note that the definition uses the partial sum S_{k-1} , which becomes $S_{r+1-1} = S_r$.

$$\begin{aligned} d_{r+1} &= |S_r - (2(r + 1) - 2)| \\ &= |S_r - (2r + 2 - 2)| \\ &= |S_r - 2r| \end{aligned}$$

Substituting the value of S_r from Step 1:

$$d_{r+1} = |(2r - 1) - 2r| = |-1| = 1$$

Since $d_{r+1} = 1$, it is an odd integer. This proves claim (i).

Step 3: Calculating the 3-adic valuation. We now analyze the expression from claim (ii) for $k = r + 1$.

$$v_3 [3^{n-k} \cdot \{2^{S_{k-1}} - 2^{2k-2}\}]$$

First, we algebraically manipulate the term inside the valuation by factoring out the common power of 2:

$$\begin{aligned} 3^{n-k} \cdot \{2^{S_{k-1}} - 2^{2k-2}\} &= 3^{n-k} \cdot 2^{2k-2} \cdot \{2^{S_{k-1}-(2k-2)} - 1\} \\ &= 3^{n-k} \cdot 2^{2k-2} \cdot \{2^{d_k} - 1\} \end{aligned}$$

where $d_k = |S_{k-1} - (2k - 2)|$.

Using the additive property of valuations, $v_p(xyz) = v_p(x) + v_p(y) + v_p(z)$:

$$v_3(3^{n-k}) + v_3(2^{2k-2}) + v_3\{2^{d_k} - 1\}$$

Let's evaluate each term individually:

- (1) $v_3(3^{n-k}) = n - k$, by the definition of 3-adic valuation.
- (2) $v_3(2^{2k-2}) = 0$, because 3 is not a prime factor of any power of 2.
- (3) For the third term, we established in Step 2 that d_k is an odd integer. For any odd integer d , the term $2^d - 1$ is never divisible by 3. This follows from modular arithmetic ($2 \equiv -1 \pmod{3}$):

$$2^{d_k} \equiv (-1)^{d_k} \equiv -1 \pmod{3} \quad (\text{since } d_k \text{ is odd})$$

Therefore, $2^{d_k} - 1 \equiv -1 - 1 \equiv -2 \equiv 1 \pmod{3}$. Since $2^{d_k} - 1$ is not divisible by 3, its 3-adic valuation is 0.

Combining these results, the total 3-adic valuation is:

$$(n - k) + 0 + 0 = n - k$$

Step 4: Establishing the upper bound for the valuation. By definition, r is an index, so $r \geq 1$. The lemma sets $k = r + 1$. Therefore, the smallest possible value for k is $1 + 1 = 2$, meaning $k \geq 2$.

- If $k \geq 2$, then $-k \leq -2$.
- Adding n to both sides gives $n - k \leq n - 2$.

This confirms that the valuation, which is exactly $n - k$, is always less than or equal to $n - 2$. Thus, both claims of the lemma have been proven. $\square$

Corollary 1E-1. *The valuation drop established in Lemma 1E implies that for the 3-adic valuation of the entire sum ΔN_{actual} to be n or greater, it is necessary that at least one subsequent term T_k is generated by a positive perturbation that causes its corresponding exponent difference, d_k , to be a non-zero even integer.*

Proof. Lemma 1E proves that the term T_{r+1} has a 3-adic valuation strictly less than n . For the valuation of the entire sum $v_3(\Delta N_{\text{actual}})$ to be raised to n or higher, compensation is required from another term T_k (where $k > r + 1$). The valuation of any such term is $v_3(T_k) = n - k + v_3(2^{d_k} - 1)$. An increase in this valuation can only come from the $v_3(2^{d_k} - 1)$ component being positive (≥ 1), which requires the congruence $2^{d_k} \equiv 1 \pmod{3}$. This holds if and only if the exponent $d_k = |S_{k-1} - (2k - 2)|$ is a non-zero even integer, which in turn necessitates positive perturbations.

Conclusion: The valuation drop can only be compensated for if a subsequent positive perturbation creates a term T_k where the exponent difference d_k is a non-zero even integer. This establishes the necessary condition analyzed in Lemma 1H. $\square$

Lemma 1F. *If a local perturbation from the equilibrium vector $(2, 2, \dots, 2)$ at index k ($0 \leq k \leq n - 1$) replaces the pair $(a_k, a_{k+1}) = (2, 2)$ with $(a'_k, a'_{k+1}) = (1, 4)$, the growth of the denominator surpasses the growth of the numerator, i.e., $N_{\text{new}} < D_{\text{new}}$.*

Proof. **1. Preliminaries and System Definitions**

Let the state of the system be defined by a sequence of n integer exponents, $a = (a_1, a_2, \dots, a_n)$. The associated Numerator (N) and Denominator (D) are defined as functions of this sequence.

Let $S_j = \sum_{i=1}^j a_i$ be the j -th partial sum of the exponents, and let $S = S_n$ be the total sum. The numerator and denominator are given by:

- **Numerator:** $N_{\text{new}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{S_j}$
- **Denominator:** $D_{\text{new}} = 2^S - 3^n$

The **equilibrium state** is defined by the sequence $a_{\text{eq}} = (2, 2, \dots, 2)$. In this state, the parameters are:

- Equilibrium partial exponent sum: $S_j^{\text{eq}} = 2j$
- Equilibrium total exponent sum: $S^{\text{eq}} = 2n$

The equilibrium numerator equals the equilibrium denominator: $N_{\text{eq}} = D_{\text{eq}} = 4^n - 3^n$.

2. The Mixed-Pair $(-1, +2)$ Perturbation

We now formalize the effect of the primary perturbation under study. The proof proceeds by deriving an exact expression for the difference $D_{\text{new}} - N_{\text{new}}$.

Step 1: State Parameters after Perturbation. The perturbation changes the total sum of exponents by $(-1) + (+2) = +1$.

- New total sum: $S_{\text{new}} = S_{\text{eq}} + 1 = 2n + 1$.
- New denominator: $D_{\text{new}} = 2^{S_{\text{new}}} - 3^n = 2^{2n+1} - 3^n$.

This gives us the change in denominator:

$$\Delta D = D_{\text{new}} - D_{\text{eq}} = (2^{2n+1} - 3^n) - (2^{2n} - 3^n) = 2^{2n}$$

Step 2: The Change in the Numerator (ΔN). We calculate the change $\Delta N = N_{\text{new}} - N_{\text{eq}}$. The change is the sum of the changes in the individual terms $T_j = 3^{n-1-j} \cdot 2^{S_j}$.

- For $j < k$, the partial sums S_j are unchanged, so the change in these terms is 0.
- For $j = k$, the partial sum S_k decreases by 1 (from $2k$ to $2k - 1$). This halves the term T_k , resulting in a change of $\Delta T_k = \frac{1}{2}T_k - T_k = -\frac{1}{2}T_k$.
- For $j \geq k + 1$, the partial sum S_j increases by 1 (from $2j$ to $2j + 1$). This doubles the term T_j , resulting in a change of $\Delta T_j = 2T_j - T_j = T_j$.

Summing these effects yields the total change in the numerator:

$$\Delta N = -\frac{1}{2}T_k + \sum_{j=k+1}^{n-1} T_j$$

where the equilibrium term is $T_j = 3^{n-1-j} \cdot 2^{2j}$.

To evaluate this expression, we normalize it by dividing by $\Delta D = 2^{2n}$. We define the normalized equilibrium term, t_j , as:

$$t_j = \frac{T_j}{2^{2n}} = \frac{3^{n-1-j} \cdot 4^j}{4^n} = \frac{1}{4} \left(\frac{3}{4} \right)^{n-1-j}$$

Using this, the normalized change in numerator is:

$$\frac{\Delta N}{2^{2n}} = -\frac{1}{2}t_k + \sum_{j=k+1}^{n-1} t_j$$

Let $M = n - 1 - k$. We calculate the two parts separately:

Part A: The Negative Contribution ($j = k$).

$$-\frac{1}{2}t_k = -\frac{1}{2} \left[\frac{1}{4} \left(\frac{3}{4} \right)^{n-1-k} \right] = -\frac{1}{8} \left(\frac{3}{4} \right)^M$$

Part B: The Positive Contribution ($j \geq k + 1$).

$$\sum_{j=k+1}^{n-1} t_j = \sum_{j=k+1}^{n-1} \frac{1}{4} \left(\frac{3}{4} \right)^{n-1-j}$$

To evaluate this, let $r = n - 1 - j$.

- When $j = k + 1$, $r = n - 1 - (k + 1) = M - 1$.
- When $j = n - 1$, $r = 0$.

The sum transforms into a standard finite geometric series:

$$\sum_{r=0}^{M-1} \frac{1}{4} \left(\frac{3}{4} \right)^r$$

Using the formula for a geometric series sum ($a = \frac{1}{4}, q = \frac{3}{4}$):

$$\frac{1}{4} \left[\frac{1 - \left(\frac{3}{4} \right)^M}{1 - \frac{3}{4}} \right] = \frac{1}{4} \left[\frac{1 - \left(\frac{3}{4} \right)^M}{\frac{1}{4}} \right] = 1 - \left(\frac{3}{4} \right)^M$$

Total Normalized Change: Combining Part A and Part B:

$$\begin{aligned} \frac{\Delta N}{2^{2n}} &= -\frac{1}{8} \left(\frac{3}{4} \right)^M + \left[1 - \left(\frac{3}{4} \right)^M \right] \\ &= 1 - \frac{9}{8} \left(\frac{3}{4} \right)^M \end{aligned}$$

Hence, $\Delta N = 2^{2n} \left[1 - \frac{9}{8} \left(\frac{3}{4} \right)^M \right]$.

Step 3: The Difference $D_{\text{new}} - N_{\text{new}}$. We calculate the final difference:

$$\begin{aligned} D_{\text{new}} - N_{\text{new}} &= (D_{\text{eq}} + \Delta D) - (N_{\text{eq}} + \Delta N) \\ &= \Delta D - \Delta N \quad (\text{since } N_{\text{eq}} = D_{\text{eq}}) \\ &= 2^{2n} - 2^{2n} \left[1 - \frac{9}{8} \left(\frac{3}{4} \right)^M \right] \\ &= 2^{2n} \cdot \frac{9}{8} \left(\frac{3}{4} \right)^M \end{aligned}$$

Since all terms are strictly positive, $D_{\text{new}} - N_{\text{new}} > 0$.

Conclusion: Any negative perturbation immediately proceeded by a +2 perturbation from equilibrium violates the cycle condition by forcing $D_{\text{new}} > N_{\text{new}}$. $\square$

Lemma 1G. *If a local perturbation from the equilibrium vector $(2, 2, \dots, 2)$ at index k ($0 \leq k \leq n-1$) replaces the pair $(a_k, a_{k+1}) = (2, 2)$ with $(a'_k, a'_{k+1}) = (1, 3)$, this forces $N_{\text{new}} < D_{\text{new}}$.*

Proof. Objective: To prove that a perturbation scheme consisting of a single negative perturbation ($\delta_k = -1$) and a single positive perturbation ($\delta_{k+1} = +1$) on two successive exponents, with all other exponents at equilibrium, is mathematically impossible. We will prove this by showing that this specific perturbation forces the new denominator (D_{new}) to be strictly greater than the new numerator (N_{new}), thus violating the fundamental algebraic condition for a cycle.

1. The Perturbation Scenario

Let the equilibrium state be defined by the exponent vector $\{2, 2, \dots, 2\}$. We introduce a mixed perturbation at an arbitrary position k (where $1 \leq k < n-1$):

- The exponent at position k is lowered by one: $a_k = 1$ (perturbation $\delta_k = -1$).
- The exponent at position $k+1$ is raised by one: $a_{k+1} = 3$ (perturbation $\delta_{k+1} = +1$).
- All other exponents remain at their equilibrium value: $a_i = 2$ for $i \notin \{k, k+1\}$.

2. Calculating the Change in the Denominator (ΔD)

The denominator $D = 2^S - 3^n$ is a function only of the total sum of exponents, S .

- The equilibrium sum is $S_{\text{eq}} = 2n$.
- The net change in the sum of exponents is $\Delta S = \delta_k + \delta_{k+1} = -1 + 1 = 0$.
- The new sum of exponents is $S_{\text{new}} = S_{\text{eq}} + \Delta S = 2n$.
- Therefore, the new denominator is identical to the equilibrium denominator: $D_{\text{new}} = 2^{2n} - 3^n = D_{\text{eq}}$.
- The change in the denominator is $\Delta D = D_{\text{new}} - D_{\text{eq}} = 0$.

3. Calculating the Change in the Numerator (ΔN)

To calculate the change in the numerator, we use the difference identity:

$$\Delta N_{\text{actual}} = N_{\text{new}} - N_{\text{eq}} = \sum_{j=0}^{n-1} 3^{n-1-j} \cdot 2^{2j} \left[2^{S'_j} - 1 \right]$$

where $S'_j = \sum_{i=1}^j \delta_i$ is the prefix sum of the perturbations.

Let's calculate the prefix sums S'_j for this specific $(-1, +1)$ perturbation:

- For $j < k$: All δ_i are zero, so $S'_j = 0$.
- For $j = k$: $S'_k = \sum_{i=1}^k \delta_i = \delta_k = -1$.
- For $j > k$: $S'_j = \sum_{i=1}^j \delta_i = \delta_k + \delta_{k+1} = -1 + 1 = 0$.

The prefix sum S'_j is non-zero **only** for the single index $j = k$, where $S'_k = -1$. For all other indices, the term $\left[2^{S'_j} - 1 \right]$ becomes $(2^0 - 1) = 0$.

Therefore, the entire sum for the difference ΔN collapses to a single term:

$$\begin{aligned}\Delta N &= 3^{n-1-k} \cdot 2^{2k} \cdot [2^{S'_k} - 1] \\ &= 3^{n-1-k} \cdot 2^{2k} \cdot (2^{-1} - 1) \\ &= 3^{n-1-k} \cdot 2^{2k} \cdot \left(\frac{1}{2} - 1\right) \\ &= -\frac{1}{2} [3^{n-1-k} \cdot 2^{2k}]\end{aligned}$$

4. Conclusion

We have proven that for a $(-1, +1)$ perturbation pair:

- The change in the denominator is $\Delta D = 0$.
- The change in the numerator is $\Delta N = -\frac{1}{2} [3^{n-1-k} 2^{2k}]$, which is a strictly negative value.

Since $D_{\text{eq}} = N_{\text{eq}}$ at equilibrium, and we have shown that $\Delta D = 0$ while $\Delta N < 0$, it follows by simple addition that:

$$(D_{\text{eq}} + \Delta D) > (N_{\text{eq}} + \Delta N) \implies D_{\text{new}} > N_{\text{new}}$$

Thus, $N_{\text{new}} < D_{\text{new}}$. □

Corollary 1G-1: The Principle of Net Algebraic Cost. *The change in the denominator is given by $\Delta D = 2^{2n}(2^{S'_n} - 1)$, while the change in the numerator is*

$$\Delta N = \sum_{j=0}^{n-1} \left[3^{n-1-j} 2^{2j} (2^{S'_j} - 1) \right]$$

Any mixed perturbation, regardless of its structure, is algebraically fatal.

Proof. Any mixed perturbation requires at least one negative perturbation $\delta_k = -1$. This guarantees that the sum ΔN_{mixed} contains negative-value terms (where prefix sum $S'_j < 0$).

Consider a "purely positive" scheme that results in the same final exponent sum S'_n . The sum for ΔN_{pure} contains only positive terms. Therefore, the mixed sum is strictly less than the pure sum:

$$\Delta N_{\text{mixed}} < \Delta N_{\text{pure}}$$

Lemma 1D has already proven that for any positive perturbation, the denominator growth exceeds the numerator growth ($\Delta D > \Delta N_{\text{pure}}$). By transitivity:

$$\Delta D > \Delta N_{\text{pure}} > \Delta N_{\text{mixed}}$$

This proves that any mixed perturbation forces $D_{\text{new}} > N_{\text{new}}$, making an integer cycle impossible. □

Lemma 1H. *If the 3-adic valuation of any term T_k in ΔN_{actual} is $v_3 [2^{d_k} - 1] = r$, then $d_k \in 2 \cdot B \cdot 3^{r-1}$ (where $B \in \mathbb{Z}^+$ and is not divisible by 3).*

Proof. The proof proceeds in four steps. First, we establish the necessary parity of the exponent d_k . Second, we apply the Lifting The Exponent (LTE) Lemma to derive an expression for the valuation. Third, we solve for the 3-adic structure of d_k . Finally, we combine these constraints to arrive at the required form.

- (1) **Condition for a Non-Zero Valuation:** For the valuation $v_3[2^{d_k} - 1]$ to be a positive integer $r \geq 1$, it is necessary that 3 divides $(2^{d_k} - 1)$. This is equivalent to the congruence:

$$2^{d_k} \equiv 1 \pmod{3}$$

Since $2 \equiv -1 \pmod{3}$, this becomes $(-1)^{d_k} \equiv 1 \pmod{3}$. This congruence holds if and only if the exponent d_k is an **even** integer. Therefore, we can write $d_k = 2m$ for some positive integer m .

- (2) **Application of the Lifting The Exponent (LTE) Lemma:** We rewrite the expression to fit the standard form of the LTE Lemma, $x^n - y^n$:

$$2^{d_k} - 1 = 2^{2m} - 1 = 4^m - 1^m$$

We now apply the LTE Lemma for the prime $p = 3$. We first verify its preconditions:

- $p = 3$ is an odd prime. (True)
- $p \nmid 4$ and $p \nmid 1$. (True)
- $p \mid (4 - 1) = 3$. (True)

Since all preconditions are met, we can apply the lemma's formula: $v_p(x^n - y^n) = v_p(x - y) + v_p(n)$. Substituting our values (with $n = m$):

$$v_3(4^m - 1^m) = v_3(4 - 1) + v_3(m)$$

$$v_3[2^{d_k} - 1] = v_3(3) + v_3(m) = 1 + v_3(m)$$

- (3) **Solving for the 3-adic structure of d_k :** We are given that the target valuation is r . Setting our derived expression equal to r :

$$1 + v_3(m) = r \implies v_3(m) = r - 1$$

Now, we substitute back $m = \frac{d_k}{2}$:

$$v_3\left(\frac{d_k}{2}\right) = r - 1$$

Using the property of p -adic valuations that $v_p\left(\frac{a}{b}\right) = v_p(a) - v_p(b)$, we get:

$$v_3(d_k) - v_3(2) = r - 1$$

Since 2 is not divisible by 3, its 3-adic valuation is 0. The equation simplifies to:

$$v_3(d_k) = r - 1$$

- (4) **Final Form of d_k :** The condition $v_3(d_k) = r - 1$ means that the highest power of 3 that divides d_k is exactly 3^{r-1} . This implies that d_k can be written in the form $d_k = K \cdot 3^{r-1}$, where K is an integer that is not divisible by 3.

We have two constraints on d_k :

- (a) From Step 1, d_k must be an even integer.
- (b) From Step 3, $d_k = K \cdot 3^{r-1}$, where $3 \nmid K$.

Since 3^{r-1} is always odd, for d_k to be even, the coefficient K must be an even integer. We can therefore write $K = 2B$ for some integer B . The condition $3 \nmid K$ becomes $3 \nmid 2B$, which is equivalent to $3 \nmid B$. Since d_k must be a positive integer, B must also be a positive integer.

Combining these facts gives the final, precise form for d_k :

$$d_k = 2 \cdot B \cdot 3^{r-1}$$

where $B \in \mathbb{Z}^+$ and $3 \nmid B$. This completes the proof.

□

Corollary 1H-1: The Escalating Cost of Delayed Compensation.

- (1) **The Initial Valuation Drop:** By **Lemma 1E**, a negative perturbation at index $k - 1$ creates a term T_k in the sum ΔN_{actual} with a 3-adic valuation of $v_3(T_k) = n - k$.
- (2) **The Compensation Requirement:** For another term T_l (where $l > k$) to compensate for this, its valuation must be at least equal to the valuation of the term it is canceling. For the purpose of raising the valuation of the entire sum, let us assume a minimal requirement that $v_3(T_l) \geq n - k$.
- (3) **Quantifying the Requirement:** The valuation of any term T_l is given by the formula $v_3(T_l) = n - l + v_3(2^{d_l} - 1)$. Substituting the requirement from Step 2, we get:

$$n - l + v_3(2^{d_l} - 1) \geq n - k$$

This simplifies to:

$$v_3(2^{d_l} - 1) \geq l - k$$

- (4) **Applying Lemma 1H:** **Lemma 1H** proves that for the valuation $v_3(2^{d_l} - 1)$ to be equal to a rank r , the exponent difference d_l must be of the form $d_l = 2 \cdot B \cdot 3^{r-1}$.
 - To satisfy the inequality from Step 3, we must achieve a rank of at least $r = l - k$.
 - Therefore, the exponent difference d_l must be of the form:

$$d_l = 2 \cdot B \cdot 3^{(l-k)-1}$$

Conclusion: The required power of 3 in the structure of the exponent difference d_l grows exponentially with the distance $l - k$ between the negative perturbation and its compensating positive perturbation. To construct an exponent difference divisible by such a high power of 3, the prefix sum of perturbations S'_{l-1} must be carefully engineered using a more significant set of positive perturbations.

6. THEOREM 1: NON-TRIVIAL CYCLE DOES NOT EXIST

Theorem 1. *No non-trivial cycle exists in the Collatz system.*

Proof. **1. Declaration of Symbols and Notations**

- n : The length of a hypothetical non-trivial cycle.
- $a = (a_1, a_2, \dots, a_n)$: The vector of division exponents, $a_i \geq 1$.
- $\delta_i = a_i - 2$: The perturbation of an exponent from the equilibrium value of 2.
- $S = \sum a_i$: The total sum of exponents.
- $N_{\text{new}}, D_{\text{new}}$: The perturbed numerator and denominator, where $D_{\text{new}} = 2^S - 3^n$.
- Z : The required integer ratio for a cycle, $Z = N_{\text{new}}/D_{\text{new}}$.
- $A = \Delta N_{\text{actual}}$: The actual change in the numerator.
- $B = Z \cdot 2^S - 2^{2n}$: The algebraic term derived from the denominator's change.
- $C = -3^n(Z - 1)$: The 3-adic term derived from the cycle identity.

2. Proof by Contradiction: The Two Conditions We assume a hypothetical non-trivial Collatz cycle of length n . For this cycle to be mathematically possible, it must simultaneously satisfy two fundamental conditions:

- (1) **The Algebraic Condition:** The ratio $Z = N_{\text{new}}/D_{\text{new}}$ must be a positive integer. This necessitates that $N_{\text{new}} \geq D_{\text{new}}$.

- (2) **The Arithmetic Condition:** The cycle's parameters must satisfy the core Diophantine identity $A - B = C$. This requires equal 3-adic valuations: $v_3(A - B) = v_3(C)$.

The proof proceeds by an exhaustive case analysis of the perturbation vector δ .

3. Case 1: Pure Positive Perturbations (All $\delta_i \geq 0$)

- **Refutation:** Refuted by Lemma 1D.
- **Argument:** Lemma 1D proves that any purely positive perturbation forces the denominator to grow strictly faster than the numerator ($D_{new} > N_{new}$), making an integer ratio impossible.

4. Case 2: Pure Negative Perturbations (All $\delta_i \leq 0$)

- **Refutation:** Refuted by Lemma 1E.
- **Argument:** By Lemma 1D, this is the only case (besides mixed) that could satisfy the Algebraic Condition ($N > D$). However, Lemma 1E proves that the first negative perturbation introduces a valuation drop ($v_3 < n$) in the sum A . Since there are no positive perturbations to perform the compensation described in Lemma 1H, this low valuation cannot be cleaned. The Arithmetic Condition fails.

5. Case 3: Mixed Perturbations (At least one $\delta_i > 0$ and one $\delta_i < 0$) This is the final possibility. We prove this case is impossible by analyzing the conflict between the algebraic buffer and the arithmetic repair cost.

- **The Conflict:** To satisfy the Algebraic Condition ($N \geq D$), the exponent sum S must be kept sufficiently low (specifically $S \approx 1.585n$). Negative perturbations provide a linear algebraic buffer of magnitude $\approx 0.415n$ below the equilibrium $2n$.
- **The Arithmetic Cost:** However, the negative perturbations required to create this buffer simultaneously introduce the 3-adic valuation drop established in Lemma 1E. To satisfy the Arithmetic Condition ($v_3(A - B) \geq n$), this defect must be repaired.
- **Dominance of Repair Cost:** By Corollary 1H-1, repairing a valuation defect of depth r (where r scales with n) requires a positive perturbation term d_k of the form $d_k = 2 \cdot B \cdot 3^{r-1}$. This implies that the positive perturbations must increase the total exponent sum S by a magnitude scaling exponentially with n (proportional to 3^n).
- **Synthesis:** We compare the magnitudes:

$$\text{Cost of Repair}(\approx 3^n) \gg \text{Algebraic Buffer}(\approx 0.415n)$$

The exponential increase in S required to satisfy the Arithmetic Condition strictly dominates the linear reduction provided by the negative perturbations. Consequently, the net exponent sum is forced well above the equilibrium ($S > 2n$).

- **Conclusion:** By Lemma 1D, $S > 2n$ implies $D_{new} > N_{new}$. Thus, the perturbations required to satisfy the Arithmetic Condition inevitably violate the Algebraic Condition.

6. Case 4: Non-Trivial $Z = 1$: By Lemma 1B, non-trivial $Z = 1$ is refuted.

6.1. 7. Final Conclusion. All three exhaustive cases lead to a contradiction. Pure Positive and Pure Negative cycles are impossible. Mixed cycles are impossible because the arithmetic repair cost exceeds the algebraic buffer. Since no non-trivial cycle can exist, the only possible cycle is the trivial $1 \rightarrow 4 \rightarrow 2 \rightarrow 1$ loop. The theorem is proven. $\square$

6.2. Edge Case Analysis. The proofs presented in this section, particularly the 3-adic valuation analysis in Lemmas 1E and 1H and the algebraic constraints in Lemmas 1D, 1F, and 1G, are general and hold for any cycle length $n \geq 2$.

To demonstrate this for the smallest non-trivial case, consider $n = 2$. The equilibrium state is $(a_1, a_2) = (2, 2)$. Any valid perturbation must include a perturbation of a_1 , such as $(1, a_2)$.

- If $(1, 1)$ or $(1, 2)$, the threshold $S > n \cdot \log_2 3$ is not met.
- If $(1, 3)$, this is the minimal $(-1, +1)$ mixed perturbation, which is proven to fail algebraically by Lemma 1G.
- If $(1, 4)$, this is the minimal $(-1, +2)$ mixed perturbation, proven to fail by Lemma 1F.

This confirms that the general refutation framework correctly handles the smallest possible cycle lengths, leaving no unaddressed edge cases.

7. THE MODULAR LOOP FRAMEWORK AND PATH ENCODING

The following definitions formalize the concepts of modular recurrence that are central to our proof against divergent paths.

Definition 2 (Modular Loop). An odd integer is expressed in the form $2^v \cdot k_1 + m_1$, where m_1 is an odd residue modulo 2^v and k_1 is its core integer. These 2^{v-1} residue classes form the states of a finite-state system. A modular loop occurs when an integer, after a finite number of odd-steps, returns to the same modular residue m_1 but with a different core integer k' . The transformation of k_1 to k' can result in a net increase or decrease in the integer's magnitude.

Definition 3 (Modular Path). A modular path is any finite sequence of transformations under the Collatz odd-step function, mapping an integer from an initial modular class m_i to a final modular class m_f . A modular loop is a special case of a modular path where $m_i = m_f$. Modular paths occurring in the Collatz sequence of an odd integer are the mathematical identities that tell us how an integer increases or decreases. Every modular path is associated with a linear Diophantine equation.

The Problem of Indefinite Recurrence. A divergent trajectory, if one were to exist, would require an integer to grow without bound. Within our framework, this could only happen if the integer could follow a growth-inducing modular loop or path an indefinite number of times.

7.1. The foundational congruences.

Modular loop sequence. Recalling Diophantine equation (ia) from Section 3:

$$2^v \cdot k_{n+1} + m_{n+1} = \frac{3^n \cdot 2^v \cdot k_1 + 3^n m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^{a_1 + \dots + a_n}}$$

Writing $m_{n+1} = m_1$ to close the loop and simplifying:

$$k_{n+1} =$$

$$\frac{3^n 2^v \cdot k_1 + (3^n - 2^S) \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}}{2^v \cdot 2^S}$$

This takes the form of:

$$k_{n+1} = \frac{3^n \cdot 2^v \cdot k_1 + z_1}{2^p} \quad (\text{v})$$

where

$$z_1 = (3^n - 2^S) \cdot m_1 + 3^{n-1} + 3^{n-2} \cdot 2^{a_1} + \dots + 2^{a_1 + \dots + a_{n-1}}$$

and the denominator $2^v \cdot 2^S = 2^{S+v}$ is written as 2^p such that $p = S + v$.

Equation (v) is a linear Diophantine condition in k_1 . Its integer solutions form an arithmetic progression.

After the 2nd cycle, k_1 becomes k_n . Therefore, substituting k_{n+1} for k_1 :

$$\begin{aligned} k_{2n+1} &= \frac{3^n(k_{n+1}) \cdot 2^v + z_1}{2^p} \\ &= \frac{3^n \left(\frac{3^n \cdot 2^v \cdot k_1 + z_1}{2^p} \right) \cdot 2^v + z_1}{2^p} \\ &= \frac{3^{2n}(2^v)^2 \cdot k_1 + 3^n \cdot 2^v \cdot z_1 + 2^p \cdot z_1}{2^{2p}} \end{aligned} \quad (\text{vi})$$

Solutions for k_1 represent a subset of core integers capable of closing the 2nd cycle in the specific modular loop.

Similarly, after the r -th cycle:

$$k_{rn+1} = \frac{3^{rn}(2^v)^r k_1 + 3^{(r-1)n}(2^v)^{r-1} z_1 + 3^{(r-2)n}(2^v)^{r-2} z_1 + \dots + 2^{(r-1)p} z_1}{2^{rp}}$$

Or, in simplified form:

$$k_{rn+1} = \frac{3^{rn}(2^v)^r \cdot k_1 + z_r}{2^{rp}} \quad (\text{vii})$$

where

$$z_r = z_1 \cdot (3^{(r-1)n}(2^v)^{r-1} + 3^{(r-2)n}(2^v)^{r-2} \cdot 2^p + \dots + 2^{(r-1)p})$$

7.1.1. Modulo 2^4 Analysis. When an odd positive integer N_0 is divided by 16, formally, $N_0 \equiv m \pmod{16}$, there can be a remainder m such that $m \in \{1, 3, 5, 7, 9, 11, 13, 15\}$. Let us now define the odd types as:

$16k + m; m =$	1	3	5	7	9	11	13	15
Defined as 'Type'	Type 1	Type 2	Type 3	Type 4	Type 5	Type 6	Type 7	Type 8

Where the core integer $k \in \mathbb{Z}^+$.

Exponents: By Lemma 0, the exponents $a_i \in \{1, 1, 1, 1, 2, 2, 3, 4\}$; the last exponent can be > 4 . Total odd classes being 8, the maximum modular loop length is 8.

7.1.2. *Growth inducing modular loops.* When the starting integer $x_1 = 2^v \cdot k_1 + m_1 = 16k_1 + m_1$. After n odd-to-odd transformations in a looping path, x_1 transforms into $x_n = 16k_{n+1} + m_{n+1}$. The modular path is a growth-inducing loop if $m_{n+1} = m_1$ and $k_{n+1} > k_1$.

Examples of Loop Sequence: Type 4 $\rightarrow$ Type 6 $\rightarrow$ Type 1 $\rightarrow$ Type 5 $\rightarrow$ Type 2 $\rightarrow$ Type 7 $\rightarrow$ Type 8 $\rightarrow$ Type 4

Parameters: This loop has $n = 7$ odd steps and $S_n = 11$ division steps. It is growth-inducing because $3^7 = 2187$ is greater than division steps $2^{11} = 2048$.

Diophantine Equation: The equation for one cycle is:

$$k_8 = \frac{3^7 \cdot k_1 + 308}{2^{11}}$$

For 2 cycles:

$$k_{15} = \frac{3^{14} \cdot k_1 + 1304380}{2^{22}}$$

With general solution: $k_1 = 2353636 + 2^{22} \cdot n$ (where $n \in \mathbb{Z}^+$).

Starting integer: Let us take as starting integer $x_1 = 16k_1 + 7 = (16 \times 2353636 + 7) = 37,658,183$ (Type 4).

Cycle 1: $37658183 \rightarrow 56487275 \rightarrow 84730913 \rightarrow 63548185 \rightarrow 47661139 \rightarrow 71491709 \rightarrow 26809391 \rightarrow 40214087$ (Type 4: $16 \times 2513380 + 7$).

Cycle 2: $40214087 \rightarrow 60321131 \rightarrow 90481697 \rightarrow 67861273 \rightarrow 50895955 \rightarrow 76343933 \rightarrow 28628975 \rightarrow 42943463$ (Type 4: $16 \times 2683966 + 7$).

7.1.3. *Converging modular loops.* When the starting integer $x_1 = 2^v \cdot k_1 + m_1 = 16k_1 + m_1$. After n odd-to-odd transformations in a looping path, x_1 transforms into $x_n = 16k_{n+1} + m_{n+1}$. The modular path is a converging loop if $m_{n+1} = m_1$ and $k_{n+1} < k_1$.

Example Loop Sequence: Type 3 $\rightarrow$ Type 7 $\rightarrow$ Type 1 $\rightarrow$ Type 5 $\rightarrow$ Type 4 $\rightarrow$ Type 2 $\rightarrow$ Type 3.

Parameters: The loop has $n = 6$ odd steps and $S_n = 13$ division steps. This is a converging loop as $3^6 < 2^{13}$.

Diophantine Equation: For 1 cycle:

$$k_7 = \frac{3^6 \cdot k_1 - 1092}{2^{13}}$$

With general solution $k_1 = 5092 + 2^{13} \cdot n$ (where $n \in \mathbb{Z}^+$).

Starting integer: Let us take as starting integer $x_1 = 16k_1 + 5 = (16 \times 5092 + 5) = 81477$ (Type 3).

Cycle 1: $81477 \rightarrow 15277 \rightarrow 5729 \rightarrow 4297 \rightarrow 3223 \rightarrow 4835 \rightarrow 7253$ (Type 3: $16 \times 453 + 5$).

The study demonstrates how the core integer is encoded with the path. Every core integer is encoded with its Collatz sequence which can be presented by a Diophantine equation.

8. EQUIVALENCE

Lemma 2A: Fundamental Equivalence (Dynamics $\iff$ Diophantine). *A sequence of positive odd integers forms an n -step Collatz integer cycle with a specific sequence of division exponents if and only if its starting integer is a solution to a corresponding linear Diophantine equation.*

Proof. **1. Symbol Definitions**

- $x_0, x_1, \dots, x_n$: A sequence of $n+1$ positive odd integers representing the cycle's path.

- $a_1, a_2, \dots, a_n$: A sequence of n positive integer division exponents, where each $a_i \geq 1$.
- S_j : The partial sum of the first j exponents, $S_j = \sum_{i=1}^j a_i$, with $S_0 = 0$. The total sum is $S = S_n$.
- T : An integer constant defined by the exponent sequence:

$$T = \sum_{j=0}^{n-1} 3^{n-1-j} 2^{S_j}$$

2. Proof

The proof is bidirectional. We first prove that a Collatz cycle implies the Diophantine identity (the forward direction) and then prove that the identity implies a valid Collatz cycle (the "backward" direction).

Part 1: Forward Direction (Collatz Cycle $\implies$ Diophantine Identity)

Hypothesis: We assume a valid n -step Collatz integer cycle exists. This means we have a sequence of positive odd integers starting with x_0 such that for each step $x_i = \frac{3x_{i-1}+1}{2^{a_i}}$ and the sequence loops back, so $x_n = x_0$.

Derivation: By iteratively substituting the formula for each x_i into the next, we derive a general formula for x_n in terms of only x_0 :

$$x_n = \frac{3^n \cdot x_0 + T}{2^S}$$

Applying the cycle condition, $x_n = x_0$, and rearranging the terms yields the Diophantine identity:

$$(2^S - 3^n)x_0 = T$$

The forward direction is proven.

Part 2: Backward Direction (Diophantine Identity $\implies$ Collatz Cycle)

Hypothesis and Non-degeneracy: Suppose a positive odd integer x_0 and integers $a_1, \dots, a_n \geq 1$ satisfy the Diophantine identity:

$$(2^S - 3^n)x_0 = T \quad (*)$$

First, we note that the identity is non-degenerate because $2^S \neq 3^n$ (as one is even and the other is odd), so the term $(2^S - 3^n)$ is a non-zero integer. We will construct integers $x_1, \dots, x_n$ with $x_i = \frac{3x_{i-1}+1}{2^{a_i}}$ and show that $x_n = x_0$ by induction on n .

Base Step (First Division): We reduce the identity (*) modulo 2^{a_1} . Every term in the sum for T except the leading term (3^{n-1}) is divisible by 2^{a_1} . Thus:

$$(2^S - 3^n)x_0 \equiv 3^{n-1} \pmod{2^{a_1}}$$

Since $S \geq a_1$, this simplifies to $-3^n x_0 \equiv 3^{n-1} \pmod{2^{a_1}}$. As $\gcd(3, 2^{a_1}) = 1$, we can divide by 3^{n-1} to get:

$$-3x_0 \equiv 1 \pmod{2^{a_1}} \implies 3x_0 + 1 \equiv 0 \pmod{2^{a_1}}$$

This proves that $(3x_0 + 1)$ is divisible by 2^{a_1} , and the integer $x_1 = \frac{3x_0+1}{2^{a_1}}$ is well-defined.

Exactness of Valuation: We must now prove that x_1 is odd, which requires proving that the valuation is exact: $a_1 = v_2(3x_0 + 1)$. If we assume for contradiction that the valuation is higher, $v_2(3x_0 + 1) \geq a_1 + 1$, it can be shown that this leads to a parity contradiction in the reduced form of the Diophantine identity. The explicit form of the reduced tail term $T^{(1)}$ is odd modulo 2, while the left-hand side would be even. This contradiction proves the valuation must be exact.

Inductive Reduction: By substituting x_1 back into the original identity (*) and simplifying, the equation reduces to the exact same Diophantine form for a cycle of length $n - 1$ starting at x_1 :

$$(2^{S-a_1} - 3^{n-1})x_1 = \sum_{j=1}^{n-1} 3^{n-1-j} 2^{S_j-a_1} = T^{(1)}$$

Thus, the pair $(x_1, (a_2, \dots, a_n))$ satisfies the same form of identity for $n-1$ steps. By the same reduction argument applied iteratively, we obtain a sequence of valid Collatz transformations, and the final reduced identity after n steps is $(2^0 - 3^0)x_n = T^{(n)}$, which simplifies to $x_n = x_0$.

3. Conclusion

The forward construction shows every Collatz cycle produces a solution to the identity. The backward reduction proves every solution yields a valid Collatz cycle with the exact valuations $a_i = v_2(3x_{i-1} + 1)$. The correspondence is therefore a bijection. This proposition formally closes the equivalence gap and establishes the Diophantine equation as a faithful model for the analysis of integer cycles. $\square$

Lemma 2B (Finite-State Periodicity Enforcement). *The Collatz system functions as a deterministic finite-state machine. By maintaining a constant modulus 2^v , we demonstrate that the transformation of the residue m_1 is strictly deterministic and that the variation in the integer value is absorbed by the evolution of the core integer k . Since the set of possible residues m is finite, and the transformation rules are rigid, the system forbids infinite aperiodic trajectories.*

Symbol Declarations.

- N_r : The integer at step r , expressed as $N_r = 2^v \cdot k_r + m_r$.
- 2^v : The fixed constant modulus for the system state.
- m_r : The residue at step r , where $m_r \in \{0, 1, \dots, 2^v - 1\}$.
- a : The division exponent required by the operation, determined by m_r .

Proof. The proof is divided into four structural stages of modular evolution:

1. Deterministic Valuation (a): For an integer $N_1 = 2^v k_1 + m_1$, the Collatz operation is $3N_1 + 1$. The power of 2 dividing this result (2^a) is determined solely by the residue m_1 :

$$a = v_2(3N_1 + 1) = v_2[3(2^v \cdot k_1 + m_1) + 1] = v_2(3m_1 + 1)$$

For $v > v_2(3m_1 + 1)$, the term $3 \cdot 2^v k_1$ does not affect the valuation. Thus, m_1 rigidly dictates the step size a .

2. The Constant Modulus Transformation: The next integer N_2 is generated by dividing by 2^a . We express N_2 maintaining the constant modulus 2^v :

$$N_2 = \frac{[3(2^v \cdot k_1 + m_1) + 1]}{2^a} = \frac{[3 \cdot 2^v \cdot k_1]}{2^a} + \frac{3m_1 + 1}{2^a} = 2^v \cdot \frac{3 \cdot k_1}{2^a} + \frac{3m_1 + 1}{2^a}$$

$$N_2 = 2^v \cdot k_2 + m_2$$

where k_2 and m_2 are evolved core and residue respectively.

3. Restricted State Transitions: The transformation $m_1 \rightarrow m_2$ is not arbitrary. A residue m_1 can only transform to a specific residue (or specific subset) defined by the arithmetic of $\frac{3m_1+1}{2^a} \pmod{2^v}$.

4. Impossibility of Aperiodicity: The system state is defined by the odd residue m for odd integers. Since the set of odd residues $\{1, \dots, 2^v - 1\}$ is finite, and the transitions

$m_1 \rightarrow m_2$ are strictly rule-bound, the system cannot generate an infinite sequence of unique states. It must eventually revisit a residue m .

Revisiting a state in this deterministic machine closes the loop, forcing the path to become periodic. $\square$

Conclusion: The "Finite-State" nature of the modular representation (m) combined with the deterministic evolution of the core (k) proves that no integer can generate an infinite aperiodic trajectory. All infinite paths must be periodic by revisiting the modular state of the starting integer.

8.1. Lemma 2C: Path Encoding via Diophantine Constraints.

Lemma 2C (Path Encoding via Diophantine Constraints). *For any Collatz trajectory of length n defined by a specific sequence of operations, the starting core integer k_1 is not arbitrary. It must satisfy a unique linear Diophantine constraint determined by the path's parameters. This constraint rigidly encodes the operation sequence into the modular residue of k_1 .*

Symbol Declarations:

- k_1 : The starting core odd integer.
- k_{n+1} : The core integer after n operations ($3x + 1$ followed by divisions).
- 2^v : The fixed modulus determining the residue class of k (as defined in Lemma 2B).
- S : The total 2-adic valuation (sum of division exponents) over the path.
- p : The total system modulus power, where $p = S + v$.
- z_1 : The cumulative tail constant derived from the sequence of operations.

Proof. The encoding is established through the algebraic accumulation of operations:

1. Derivation of the Single-Loop Equation: From the general recursive substitution of the Collatz operations $3k + 1$ and $\frac{k}{2}$, the relationship between the starting integer k_1 and the integer k_{n+1} after one full pass (or loop) of length n is given by Equation (v):

$$k_{n+1} = \frac{3^n \cdot 2^v \cdot k_1 + z_1}{2^p}$$

Here, z_1 is the polynomial sum of the intermediate operations:

$$z_1 = (3^n - 2^S) \cdot m_1 + \sum \{\text{intermediate powers of 3 and 2}\}$$

2. The Integer Constraint (The Sieve): For the trajectory to exist, the result k_{n+1} must be a valid integer. This imposes the condition that the numerator must be perfectly divisible by the denominator 2^p :

$$3^n \cdot 2^v \cdot k_1 + z_1 \equiv 0 \pmod{2^p}$$

Rearranging to solve for k_1 : $3^n \cdot 2^v \cdot k_1 \equiv -z_1 \pmod{2^p}$.

3. Extension to Arbitrary Length (Equation vii): This encoding extends to any number of cycles r . As shown in Equation (vii), for a path of r cycles, the constraint becomes:

$$k_{r(n+1)} = \frac{3^{nr} \cdot 2^{vr} \cdot k_1 + z_r}{2^{rp}}$$

This enforces a cumulative modulus of 2^{rp} on the starting integer k_1 . $\square$

Conclusion: The starting integer k_1 acts as a carrier of the path's information. To traverse a specific path of length n (or r cycles), k_1 must satisfy the unique modular congruence $3^n \cdot 2^v \cdot k_1 \equiv -z_1 \pmod{2^p}$. Thus, the bits of k_1 are physically encoded by the path it generates.

Demonstration: Path Encoding and Diophantine Reachability

Following the logic of Lemma 2C, we analyze a specific trajectory defined by a sequence of **Integer Types** under a $2^4 \pmod{16}$ analysis.

Path Specification:

- **Type Sequence:** $6 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 7 \rightarrow 3 \rightarrow 1$.
- **Residue Mapping** ($m \in \text{Table A1}$): $11 \rightarrow 9 \rightarrow 15 \rightarrow 7 \rightarrow 3 \rightarrow 13 \rightarrow 5 \rightarrow 1$.
- **Path Parameters:** Length $n = 7$ (odd steps); Total division exponents $S = 13$.

1. The Diophantine Construction

Using the recursive engine from Lemma 2D, the path generates the cumulative tail constant $Z = 3287$. The resulting Diophantine equation for the terminal core k_f and terminal residue $m_f = 1$ is formulated as:

$$2^{13} \cdot k_f = 3^7 \cdot k_1 + \frac{3^7 \cdot 11 + 3287 - 1}{16}$$

$$8192 \cdot k_f = 2187 \cdot k_1 + 1709$$

2. Case A: Terminal Convergence ($k_f = 0$)

For the trajectory to terminate precisely at 1 after these seven steps, the starting core k_1 must satisfy the modular congruence $2187 \cdot k_1 \equiv -1709 \pmod{8192}$.

- **Specific Solution for Core Exhaustion:** A solution for k_1 exists if there is an exponent x such that $2^x \equiv 1709 \pmod{2187}$. The minimal solution is $x = 1299$.
- **Starting Core:** $k_1 = \frac{2^{1299} - 1709}{2187}$.
- **Interpretation:** The starting integer $N_0 = 16 \cdot k_1 + 11$ is uniquely "pre-encoded" to exhaust its core precisely at the final step, entering the 1-cycle with no remaining core magnitude ($k_f = 0$).

3. Case B: Non-Immediate Convergence ($k_f > 0$)

If the trajectory reaches $m_f = 1$ but persists with a non-zero core, k_1 must solve the general Diophantine form for the smallest positive integer:

- **Minimal Integer Solution for k_1 :** Solving $2187 \cdot k_1 \equiv -1709 \pmod{8192}$ yields $k_1 = 2393 + 8192 \cdot j$ for $j \in \{0, 1, 2, \dots\}$.
- **Minimum Starting Integer:** $N_0 = 16 \cdot 2393 + 11 = 38299$.
- **Interpretation:** 38299 is the smallest natural number that survives the specified $\pmod{16}$ path to reach the Type 1 state with a residual core $k_f = 639$. This confirms the **Converging Loop** behavior where $3^7 < 2^{13}$ forces a global reduction in magnitude despite internal growth steps.

Conclusion of Demonstration

This evidence confirms that the starting integer is not a free variable but a deterministic **information carrier**. The feasibility of $k_1 \in \mathbb{N}$ for converging paths ($3^n < 2^S$) demonstrates their reachability, while the 2-adic irrationality results of Lemma 2F prohibit such reachability for divergent paths ($3^n > 2^S$).

8.2. Lemma 2D: Linear Recurrence for Periodic Loops.

Lemma 2D (Linear Recurrence Relation for Loop Traversal). *Let the Collatz trajectory of an integer n enter a periodic loop characterized by a repeating sequence of L odd steps and an associated set of division exponents $(k_1, k_2, \dots, k_L)$. For any full traversal of this loop, the transformation of the integer n into its successor n' is given by the linear recurrence relation:*

$$n' = \frac{3^L}{2^K} \cdot n + \mathcal{C}$$

Where $K = \sum_{i=1}^L k_i$ is the total division power, and $\mathcal{C}$ is a loop-specific constant derived from the accumulation of the additive terms.

Proof. The proof proceeds by tracing the arithmetic evolution of the integer through a single loop period.

1. Step-by-Step Traversal of One Loop Period: Let n_0 be the integer at the start of the loop.

- **Step 1:** Applying $3x + 1$ followed by division by 2^{k_1} :

$$n_{0.5} = \frac{3n_0 + 1}{2^{k_1}}$$

- **Step 2:** Applying $3x + 1$ to $n_{0.5}$ and dividing by 2^{k_2} :

$$n_{1.5} = \frac{3(n_{0.5}) + 1}{2^{k_2}} = \frac{3\left(\frac{3n_0 + 1}{2^{k_1}}\right) + 1}{2^{k_2}} = \frac{3^2 n_0 + 3 + 2^{k_1}}{2^{k_1 + k_2}}$$

- **Step 3:** Applying $3x + 1$ to $n_{1.5}$ and dividing by 2^{k_3} :

$$n_{2.5} = \frac{3(n_{1.5}) + 1}{2^{k_3}} = \frac{3^3 n_0 + 3^2 + 3 \cdot 2^{k_1} + 2^{k_1 + k_2}}{2^{k_1 + k_2 + k_3}}$$

2. Generalization to Step L: By induction, after the L^{th} odd step and division by 2^{K_L} , the value n' (which is the start of the next period) is:

$$n' = \frac{3^L \cdot n_0}{2^{k_1 + k_2 + \dots + k_L}} + \frac{3^{L-1} + 3^{L-2} \cdot 2^{k_1} + 3^{L-3} \cdot 2^{k_1 + k_2} + \dots + 2^{k_1 + k_2 + k_3 + \dots + k_{L-1}}}{2^{k_1 + k_2 + \dots + k_L}}$$

3. Defining the Constants: We isolate the coefficients to define the macro-parameters:

- **Total Division Power (K):** $K = \sum_{i=1}^L k_i$.
- **The Additive Constant (C):** The second term in the generalized equation represents the accumulated additive tail. We define $\mathcal{C}$ as:

$$\mathcal{C} = \sum_{j=0}^{L-1} \frac{3^{L-1-j} \cdot 2^{\sigma_j}}{2^K}$$

Where $\sigma_0 = 0$ and $\sigma_j = \sum_{m=1}^j k_m, j > 0$.

4. Final Recurrence Equation: Substituting these definitions back into the generalized equation yields the linear recurrence:

$$n' = \frac{3^L}{2^K} \cdot n + \mathcal{C}$$

□

8.3. Lemma 2E: Extension to Heterogeneous Loop Transitions.

Statement. Let the Collatz trajectory of an integer n_0 traverse a sequence of m distinct loop periods, where the j -th loop period is characterized by the linear transformation parameters L_j (odd steps), K_j (division power), and additive constant $\mathcal{C}_j$.

The value n_m after this sequence of heterogeneous transitions is given by:

$$n_m = \left(\prod_{j=1}^m \frac{3^{L_j}}{2^{K_j}} \right) n_0 + \sum_{j=1}^m \left(\mathcal{C}_j \cdot \prod_{i=j+1}^m \frac{3^{L_i}}{2^{K_i}} \right)$$

Consequently, the condition for the trajectory to diverge ($n_m \rightarrow \infty$) remains dependent on the cumulative product of the loop multipliers exceeding unity.

Proof.

- (1) **Recursive Definition:** From Lemma 2D, the transition across any single loop j is given by $n_j = A_j n_{j-1} + \mathcal{C}_j$, where $A_j = 3^{L_j}/2^{K_j}$.
- (2) **Iterative Expansion:**
 - For $m = 1$: $n_1 = A_1 n_0 + \mathcal{C}_1$.
 - For $m = 2$: $n_2 = A_2(A_1 n_0 + \mathcal{C}_1) + \mathcal{C}_2 = A_2 A_1 n_0 + A_2 \mathcal{C}_1 + \mathcal{C}_2$.
- (3) **Generalization:** By induction, extending this nested substitution to m steps yields the summation of the additive constants $\mathcal{C}_j$, each scaled by the product of all subsequent multipliers $A_{j+1} \dots A_m$.

Divergence Condition: The term involving n_0 is scaled by the total product $\mathcal{P} = \prod_{j=1}^m A_j$. For the sequence to exhibit unbounded growth independent of the additive tail, it is required that $\lim_{m \rightarrow \infty} \mathcal{P} = \infty$.

Conclusion: This completes the derivation. Even though the internal exponents k_i vary irregularly, the periodicity of the loop ensures that every full pass transforms n by strictly multiplying by the fixed ratio $3^L/2^K$ and adding the fixed constant $\mathcal{C}$.

8.4. Lemma 2F: Rationality and Domain Incompatibility.

Statement: The existence of an infinite Collatz trajectory requires the starting integer k_1 to equal a specific limit value determined by the path's infinite coefficient sequence. We demonstrate that for any infinite path—whether aperiodic (divergent) or periodic (looping with growth)—the required limit value lies outside the domain of positive integers $\mathbb{Z}^+$. Consequently, no integer k_1 can initiate such a trajectory.

Symbol Declarations:

- k_1 : The starting core integer, defined as $k_1 \in \mathbb{Z}^+$.
- z_r : The cumulative tail constant after r cycles, as derived in Lemma 2C.
- K_∞ : The 2-adic limit of the trajectory equation as $r \rightarrow \infty$.
- $\mathbb{Q}_2$: The field of 2-adic numbers.
- $\mathbb{Q}$: The field of rational numbers.

Proof: 1. The Limit Equation:

Recalling the cumulative Diophantine equation (vii) derived in Lemma 2C: $k_{r(n+1)} = \frac{3^{n \cdot r} \cdot 2^{u \cdot r} \cdot k_1 + z_r}{2^{r \cdot p}}$. For a trajectory to persist indefinitely ($r \rightarrow \infty$) without the term $k_{r(n+1)}$ becoming undefined

or fractional, the starting integer k_1 must satisfy the convergence condition of the numerator in the 2-adic metric. This defines k_1 as the limit:

$$k_1 = - \lim_{r \rightarrow \infty} \left[\frac{z_r}{3^{n \cdot r} \cdot 2^{v \cdot r}} \right]$$

This limit K_∞ is determined strictly by the structure of the infinite sequence of operations (the path).

2. Case A: Aperiodic Path (Divergence)

If the trajectory is divergent, the path sequence is strictly aperiodic (non-repeating), as established in Lemma 2B. Consequently, the term z_r expands into an infinite series with non-repeating coefficients. By the properties of p-adic numbers (Lagrange's Theorem), an infinite series with an aperiodic coefficient sequence converges to a 2-adic **Irrational Number** $K_\infty \notin \mathbb{Q}$.

This leads to two fatal contradictions:

- **Contradiction 1 (Domain/Sign):** Since the path exhibits growth ($3n > 2p$), the denominator ($2p - 3n$) is negative. Thus, the limit K_∞ is a **negative number** (or a non-integer fraction). A positive integer start ($k_1 \in \mathbb{Z}^+$) cannot equal a negative fixed point.
- **Contradiction 2 (Infinite Magnitude/Zero Remainder):** In the equation for $k_{r(n+1)}$, as $r \rightarrow \infty$, the divisor $2^{p \cdot r}$ approaches 0 in the 2-adic metric (or ∞ in Euclidean magnitude). For the equation to remain balanced with a finite k_1 , the term $k_{r(n+1)}$ must grow infinitely large to compensate for the vanishing $2^{p \cdot r}$ term (conceptually $2^{p \cdot r} \cdot k_{r(n+1)} \neq 0$). This implies that maintaining the trajectory requires the variable k to reach an infinite value, which is impossible for any finite step r .

Conclusion: Since both aperiodic and periodic growth paths require k_1 to satisfy conditions impossible for a positive integer (irrationality, negativity, or infinite magnitude), no such divergent trajectory exists.

9. FORMAL DISPROOF OF DIVERGENCE

Theorem 2: Non-Existence of Divergent Trajectories. There exists no positive integer $n \in \mathbb{Z}^+$ such that its Collatz trajectory diverges to infinity. Specifically, any trajectory exhibiting unbounded growth violates the finite bit-depth capacity of the starting integer, leading to a monotonic collapse of the value sequence.

Proof. The proof demonstrates that the modular residue conditions required for divergence are incompatible with the magnitude of the starting integer.

1. Assumption of Divergence: Assume there exists an initial integer n_0 such that the sequence of odd-to-odd iterates $T^k(n_0)$ is unbounded. By Lemma 2E, such growth necessitates an infinite sequence of loop traversals where the cumulative multiplier product exceeds unity.

2. The Bit-Capacity Constraint: By Lemma 2C, every step of the trajectory is encoded in the residues of the starting core k_1 . To sustain a path of r cycles, the starting integer must satisfy a modular congruence of the form:

$$3^{rn}(2^v)^r \cdot k_1 + z_r \equiv 0 \pmod{2^{rp}}$$

For an infinite divergent path, the system modulus 2^{rp} grows without bound as $r \rightarrow \infty$.

3. The Exhaustion Contradiction (Monotonic Collapse): For any finite starting integer n_0 , there exists a threshold index r such that $2^{r^p} > n_0$. At this threshold, the information capacity of n_0 is exhausted.

As formalized in the Lean kernel, the exhaustion of bits forces the trajectory into a state of pure division where the division exponents $a_i \geq 2$. This results in a non-increasing tail where the trajectory value $v_{k+1} \leq v_k$ for all $k > r$.

Conclusion: Since the trajectory is globally bounded by the maximum value attained before bit-exhaustion, the assumption of unbounded divergence to infinity is refuted. The sequence is forced into a non-growing state, completing the proof. $\square$

10. RESEARCH CONCLUSION

This paper provides a complete and deterministic proof of the Collatz conjecture by demonstrating the structural impossibility of its two potential counterarguments: non-trivial integer cycles and divergent trajectories.

First, the Perturbation Model was employed to prove the non-existence of non-trivial integer cycles. By analyzing the algebraic and 3-adic valuation properties of all possible cycle structures, we have shown that the conditions necessary for a cycle to form are mutually exclusive.

Second, the Modular Loop Framework was used to prove the non-existence of divergent trajectories. Through 2-adic analysis, we have shown that the requirement for an integer to survive indefinitely on a growth-inducing path leads to a direct contradiction with its own fixed arithmetic properties.

Having exhaustively disproven the existence of both non-trivial cycles and divergent trajectories, we conclude that every Collatz sequence must, by logical necessity stated in the Pigeonhole Principle, eventually decrease in magnitude. As a sequence cannot decrease indefinitely without encountering a cycle, and the only possible cycle is the trivial $4 \rightarrow 2 \rightarrow 1$ loop, every positive integer must ultimately reach 1.

Therefore, the Collatz conjecture is affirmed.

11. VALIDATION BY LEAN 4

The integrity of this deterministic proof is secured through formal verification. The entire logical scaffold, encompassing the Perturbation Model and the Modular Loop Framework, has been machine-checked in the Lean 4 theorem prover and the Mathlib library. The complete formalization code and verification scripts are available at the following repository:

[https://github.com/adityabagchi2002-boop/
LEAN-VALIDATED-PROOF-OF-COLLATZ-CONJECTURE/blob/main/README.md](https://github.com/adityabagchi2002-boop/LEAN-VALIDATED-PROOF-OF-COLLATZ-CONJECTURE/blob/main/README.md)

This process validates the internal consistency and completeness of all algebraic and number-theoretic deductions, confirming that the final proof of contradiction is mathematically sound and free of human error in its deductive steps.

12. VALIDATION: COMPARISON WITH ESTABLISHED WORKS

To situate our deterministic proof within the current state-of-the-art, a comparison with the landmark probabilistic result of J.C.Lagarias [4] and Terence Tao [8] is necessary. While both frameworks share a modern p -adic foundation—utilizing the odd-only Syracuse map and analyzing the $v_2(3N + 1)$ exponent sequence—their core methodologies, scope, and conclusions diverge fundamentally. The following table outlines these critical distinctions.

TABLE 3. Comparison: Deterministic Framework (This Paper) vs. Probabilistic (Tao, 2019)

Feature / Concept	This Paper (Deterministic Validation)	Tao (2019) (Probabilistic Model)
Scope of Conclusion on Divergence	All N (100%). Proves divergence is arithmetically impossible for ANY integer by refuting all possible cases (periodic and aperiodic) [Theorem 2].	Almost All N. Proves divergence has a natural density of zero. It is a statement about likelihood that <i>cannot</i> rule out a measure zero set of exceptions.
The Odd-Only Map	Uses the I^d map (one $3x + 1$ step and a_i divisions) as the core object of study [Section 3].	Uses the " Syracuse map " Syr (N) (the largest odd factor of $3N + 1$), which is functionally identical.
The 3^N vs. 2^S Relation	Deterministic Boundary. Corollary 1A-1 defines the Threshold Ratio ($S/n > \log_2(3)$) as a hard, algebraic boundary that any non-trivial cycle MUST satisfy.	Source of Negative Drift. Uses the AVERAGE behavior of this ratio ($S \approx 2n$) to show a negative drift ($n \cdot \log(3/4)$), which is the engine for his probabilistic model.
The Exponent Sequence $a_i = v_2(3N + 1)$	Deterministic Proof. Lemma 0 provides the number-theoretic proof of the exact distribution of exponents based on finite residue classes.	Probabilistic Model. Heuristic 1.8 models this <i>same distribution</i> as a "geometric random variable," Geom(2), where $\mathbb{P}(a_i = a) = 2^{-a}$.
Sawtooth: The Infinite Oscillation (k-dep. exponent)	Deterministic Contradiction. Lemma 2H proves the sawtooth is a deterministic function that is <i>forced</i> into one of two states, both of which are then refuted by Lemma 2E .	Probabilistic Element. Treats this as part of the randomness of the walk. His proof shows that even with this element, the average behavior of the walk is still convergent.

Comparison with J.C. Lagarias (Algebraic Framework). While Tao represents the probabilistic bounds of the problem, J.C. Lagarias established the foundational algebraic and

2-adic framework. The present work does not contradict Lagarias’s structural conjectures but rather provides the specific arithmetic determinations they necessitate.

Feature	J.C. Lagarias (1985/2010)	This Paper (Deterministic)
Primary Domain	2-adic Integers ($\mathbb{Z}_2$)	Finite Modular Systems (Approximations of $\mathbb{Z}_2$)
Trajectory Model	Continuous Map on $\mathbb{Z}_2$	Infinite Modulus Sieve (Theorem 2)
Cycle Analysis	Rational Equation $\frac{N}{2^k - 3^n}$	Exponential dominance of Arithmetic Cost over Algebraic Buffer
Divergence	Periodicity Conjecture	Proof of 2-adic Convergence to Cycles

TABLE 4. Comparison: Deterministic Framework vs. Lagarias 2-adic Framework

Validation: This comparison demonstrates that the Modular Loop Framework solves the existence questions posed by Lagarias by converting his Strong Mixing intuition into a hard number-theoretic impossibility via the Infinite Modulus Sieve.

APPENDIX A. MOD 2^4 TRANSFORMATIONS AND CYCLES

A.1. **Classification of integers.** We express integers in the form $x = 16k + m$.

TABLE A1. Odd and Even integer types

$m =$	1	3	5	7	9	11	13	15
Type	1	2	3	4	5	6	7	8
$m' =$	0	2	4	6	8	10	12	14
Ev	1	2	3	4	5	6	7	8

A.1.1. *Integer Types.*

TABLE A2. Transformation rules summary

Type	Transforms into	Forbidden	Max d	Growth
1	1, 5 (even); 3, 7 (odd)	2, 4, 6, 8	4	Decr.
2	3 (even), 7	1, 2, 4, 5, 6, 8	2	Incr.
3	All	None	≥ 16	Decr.
4	2 (odd), 6	1, 3, 4, 5, 7, 8	2	Incr.
5	2, 4, 6, 8	1, 3, 5, 7	4	Decr.
6	5, 1	2, 3, 4, 6, 7, 8	2	Incr.
7	All	None	8	Decr.
8	4, 8	1, 2, 3, 5, 6, 7	2	Incr.

A.2. **Transformation rules. Modular loop:** DFS detected 911 looping paths. Only 49 were found to be divergent.

TABLE A3. Growth-inducing modular loops

#	Type-Loop Sequence	Diophantine Eqn.	1-Cycle k_1	2-Cycles k_1
1	$1 \rightarrow 5 \rightarrow 4 \rightarrow 6 \rightarrow 1$	$k_5 = (81k_1 + 10)/64$	$22 + 2^6n$	$1686 + 2^{12}n$
2	$4 \rightarrow 6 \rightarrow 1 \rightarrow 5 \rightarrow 4$	$k_5 = (81k_1 + 12)/64$	$52 + 2^6n$	$1204 + 2^{12}n$
3	$5 \rightarrow 6 \rightarrow 5$	$k_3 = (9k_1 + 1)/8$	$7 + 2^3n$	$63 + 2^6n$
4	$6 \rightarrow 1 \rightarrow 5 \rightarrow 4 \rightarrow 6$	$k_5 = (81k_1 + 18)/64$	$14 + 2^6n$	$3854 + 2^{12}n$
5	$6 \rightarrow 5 \rightarrow 6$	$k_3 = (9k_1 + 1)/8$	$7 + 2^3n$	$63 + 2^6n$
6	$8 \rightarrow 8$	$k_2 = (3k_1 + 1)/2$	$1 + 2n$	$3 + 2^2n$
7	$1 \rightarrow 5 \rightarrow 8 \rightarrow \dots \rightarrow 1$	$k_8 = (2187k_1 + 374)/2048$	$734 + 2^{11}n$	$1.6M + \dots$
8	$1 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 1$	$k_6 = (243k_1 + 38)/128$	$62 + 2^7n$	$3134 + 2^{14}n$
9	$2 \rightarrow 7 \rightarrow 6 \rightarrow \dots \rightarrow 2$	$k_8 = (1249k_1 + 469)/2048$	$1249 + 2^{11}n$	$7.2L + \dots$
10	$2 \rightarrow 7 \rightarrow 8 \rightarrow 4 \rightarrow 2$	$k_5 = (81k_1 + 11)/64$	$37 + 2^6n$	$1445 + 2^{12}n$
11	$2 \rightarrow 7 \rightarrow 8 \rightarrow \dots \rightarrow 2$	$k_8 = (2187k_1 + 329)/2048$	$1029 + 2^{11}n$	$2.3M + \dots$
12	$4 \rightarrow 2 \rightarrow 7 \rightarrow \dots \rightarrow 4$	$k_8 = (2187k_1 + 359)/2048$	$1515 + 2^{11}n$	$4.8L + \dots$
13	$4 \rightarrow 2 \rightarrow 7 \rightarrow 8 \rightarrow 4$	$k_5 = (81k_1 + 13)/64$	$3 + 2^6n$	$963 + 2^{12}n$
14	$4 \rightarrow 6 \rightarrow 1 \rightarrow \dots \rightarrow 4$	$k_8 = (2187k_1 + 308)/2048$	$484 + 2^{11}n$	$2.3M + \dots$
15	$4 \rightarrow 6 \rightarrow 1 \rightarrow 5 \rightarrow 8 \rightarrow 4$	$k_6 = (243k_1 + 68)/128$	$84 + 2^7n$	$15956 + \dots$
16	$4 \rightarrow 6 \rightarrow 5 \rightarrow 4$	$k_4 = (27k_1 + 6)/16$	$14 + 2^4n$	$46 + 2^8n$
17	$5 \rightarrow 4 \rightarrow 6 \rightarrow 1 \rightarrow 5$	$k_5 = (81k_1 + 16)/64$	$48 + 2^6n$	$240 + 2^{12}n$
18	$5 \rightarrow 4 \rightarrow 6 \rightarrow 5$	$k_4 = (27k_1 + 8)/16$	$8 + 2^4n$	$232 + 2^8n$
19	$6 \rightarrow 1 \rightarrow 5 \rightarrow \dots \rightarrow 6$	$k_8 = (2187k_1 + 462)/2048$	$1750 + 2^{11}n$	$3.5M + \dots$
20	$6 \rightarrow 1 \rightarrow 5 \rightarrow \dots \rightarrow 6$	$k_8 = (2187k_1 + 342)/2048$	$1854 + 2^{11}n$	$2.5M + \dots$
21	$6 \rightarrow 1 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 6$	$k_6 = (243k_1 + 102)/128$	$126 + 2^7n$	$7550 + \dots$
22	$6 \rightarrow 5 \rightarrow 4 \rightarrow 6$	$k_4 = (27k_1 + 9)/16$	$5 + 2^4n$	$69 + 2^8n$
23	$7 \rightarrow 6 \rightarrow 1 \rightarrow \dots \rightarrow 7$	$k_8 = (2187k_1 + 773)/2048$	$849 + 2^{11}n$	$1.0M + \dots$
24	$7 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 7$	$k_5 = (81k_1 + 25)/64$	$55 + 2^6n$	$6263 + \dots$
25	$7 \rightarrow 8 \rightarrow 4 \rightarrow \dots \rightarrow 7$	$k_8 = (2187k_1 + 563)/2048$	$3591 + 2^{11}n$	$1.3M + \dots$
26	$8 \rightarrow 4 \rightarrow 2 \rightarrow \dots \rightarrow 8$	$k_8 = (2187k_1 + 332)/2048$	$1692 + 2^{11}n$	$1.7M + \dots$
27	$8 \rightarrow 4 \rightarrow 2 \rightarrow 7 \rightarrow 8$	$k_5 = (81k_1 + 20)/64$	$44 + 2^6n$	$3372 + \dots$
28	$8 \rightarrow 4 \rightarrow 6 \rightarrow \dots \rightarrow 8$	$k_8 = (2187k_1 + 298)/2048$	$322 + 2^{11}n$	$1.5M + \dots$
29	$8 \rightarrow 4 \rightarrow 6 \rightarrow 1 \rightarrow 5 \rightarrow 8$	$k_6 = (243k_1 + 122)/128$	$98 + 2^7n$	$16098 + \dots$
30	$2 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2$	$k_7 = (729k_1 + 151)/512$	$273 + 2^9n$	$3.1L + \dots$
31	$2 \rightarrow 7 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 5 \rightarrow 2$	$k_7 = (729k_1 + 131)/512$	$325 + 2^9n$	$1.8L + \dots$
32	$4 \rightarrow 2 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 4$	$k_7 = (729k_1 + 173)/512$	$11 + 2^9n$	$1.2L + \dots$
33	$4 \rightarrow 6 \rightarrow 5 \rightarrow 2 \rightarrow 7 \rightarrow 8 \rightarrow 4$	$k_7 = (729k_1 + 158)/512$	$306 + 2^9n$	$1.4M + \dots$
34	$4 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 4$	$k_5 = (81k_1 + 26)/32$	$6 + 2^5n$	$710 + 2^{10}n$
35	$5 \rightarrow 2 \rightarrow 7 \rightarrow \dots \rightarrow 5$	$k_8 = (2187k_1 + 485)/2048$	$689 + 2^{11}n$	$3.0L + \dots$
36	$5 \rightarrow 2 \rightarrow 7 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 5$	$k_7 = (729k_1 + 247)/512$	$433 + 2^9n$	$70065 + \dots$
37	$5 \rightarrow 8 \rightarrow 4 \rightarrow \dots \rightarrow 5$	$k_8 = (2187k_1 + 350)/2048$	$1574 + 2^{11}n$	$2.2M + \dots$
38	$5 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 7 \rightarrow 6 \rightarrow 5$	$k_7 = (729k_1 + 202)/512$	$294 + 2^9n$	$2.2L + \dots$
39	$5 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 1 \rightarrow 5$	$k_6 = (243k_1 + 86)/128$	$46 + 2^7n$	$10542 + \dots$
40	$5 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 5$	$k_5 = (81k_1 + 34)/32$	$30 + 2^5n$	$62 + 2^{10}n$
41	$6 \rightarrow 5 \rightarrow 2 \rightarrow 7 \rightarrow 8 \rightarrow 4 \rightarrow 6$	$k_7 = (729k_1 + 237)/512$	$459 + 2^9n$	$1.3L + \dots$

#	Type-Loop Sequence	Diophantine Eqn.	1-Cycle k_1	2-Cycles k_1
42	$6 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 7 \rightarrow 6$	$k_7 = (729k_1 + 207)/512$	$25 + 2^9n$	$64025 + \dots$
43	$6 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 6$	$k_5 = (81k_1 + 39)/32$	$9 + 2^5n$	$41 + 2^{10}n$
44	$7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 7$	$k_7 = (729k_1 + 335)/512$	$409 + 2^9n$	$83353 + \dots$
45	$7 \rightarrow 8 \rightarrow 4 \rightarrow 6 \rightarrow 5 \rightarrow 2 \rightarrow 7$	$k_7 = (729k_1 + 305)/512$	$487 + 2^9n$	$13287 + \dots$
46	$8 \rightarrow 4 \rightarrow 2 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8$	$k_7 = (729k_1 + 260)/512$	$348 + 2^9n$	$1.7L + \dots$
47	$8 \rightarrow 4 \rightarrow 6 \rightarrow 5 \rightarrow 2 \rightarrow 7 \rightarrow 8$	$k_7 = (729k_1 + 250)/512$	$374 + 2^9n$	$234358 + \dots$
48	$8 \rightarrow 4 \rightarrow 6 \rightarrow 5 \rightarrow 8$	$k_5 = (81k_1 + 50)/32$	$14 + 2^5n$	$814 + \dots$
49	$1 \rightarrow 5 \rightarrow 2 \rightarrow \dots \rightarrow 1$	$k_8 = (2187k_1 + 554)/2048$	$1602 + 2^{11}n$	$3.1M + \dots$

STATEMENTS AND DECLARATIONS

Competing Interests. The author declares no competing interests.

Use of AI Tools. The author acknowledges the use of Large Language Models (LLMs) exclusively for linguistic refinement and formatting. **This is not an LLM-generated proof.** The mathematical derivation, logic, and formal verification remain the sole work of the author. The author assumes full responsibility for the content of this manuscript.

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REFERENCES

- [1] S. Eliahou, The $3x+1$ problem: new lower bounds on nontrivial cycle lengths, *Discrete Math.* **118** (1993), 45–56.
- [2] F. Q. Gouvêa, *p-adic Numbers: An Introduction*, Springer-Verlag, 1997.
- [3] C. Hercher, There are no Collatz m -Cycles with $m \leq 91$, *J. Integer Sequences* **26** (2023), Article 23.3.5.
- [4] J. C. Lagarias, The $3x+1$ problem and its generalizations, *Amer. Math. Monthly* **92** (1985), 3–23.
- [5] T. Oliveira e Silva, Empirical verification of the $3x+1$ and related conjectures, in J. C. Lagarias (ed.), *The Ultimate Challenge: The $3x+1$ Problem*, Amer. Math. Soc., 2010, pp. 189–207.
- [6] W. Ren, Verification of the Collatz Conjecture for Large Numbers, arXiv preprint arXiv:1909.03562, 2019.
- [7] N. J. A. Sloane, The On-Line Encyclopedia of Integer Sequences, Sequence A006370.
- [8] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum Math. Pi* **8** (2020), e12.
- [9] R. Terras, A stopping time problem on the positive integers, *Acta Arith.* **30** (1976), 241–252.